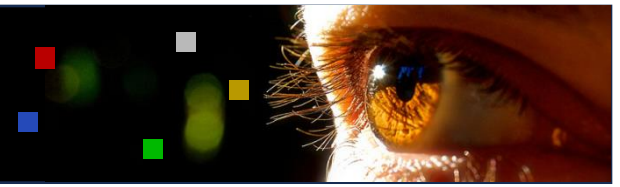
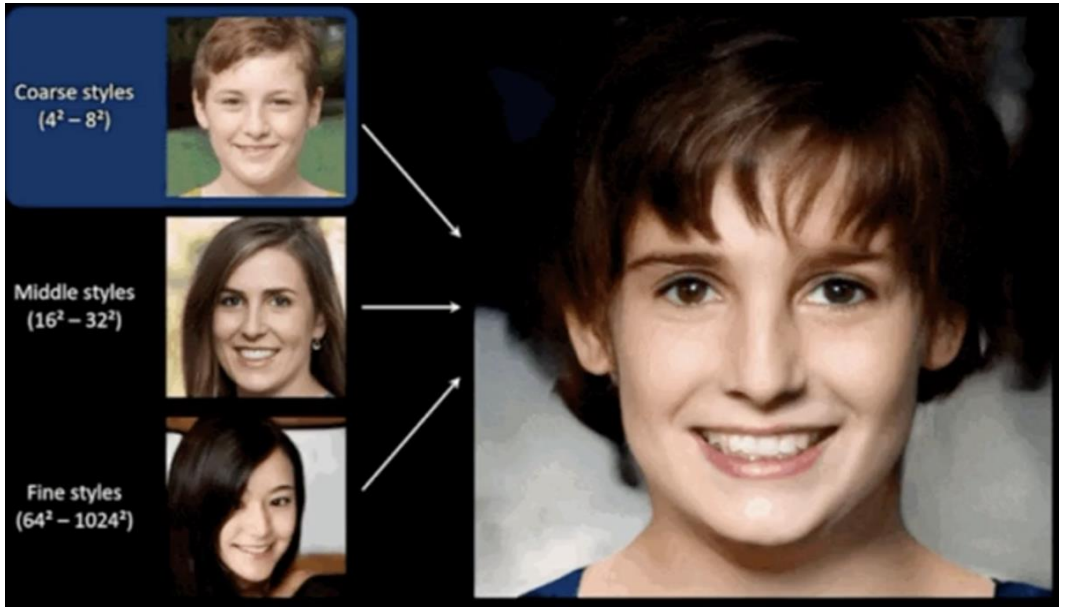
Visión por Computador

Versión 2026



GAN

(CAPÍTULO 3 – CV03F)



Domingo Mery

Ciencia de la Computación
Universidad Católica – Chile
http://domingomery.ing.uc.cl

DOI:10.1145/3422622

Generative Adversarial Networks

By Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu,
David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio

Abstract

Generative adversarial networks are a kind of artificial intelligence algorithm designed to solve the *generative modeling* problem. The goal of a generative model is to study a collection of training examples and learn the probability distribution that generated them. Generative Adversarial Networks (GANs) are then able to generate more examples from the estimated probability distribution. Generative models based on deep learning are common, but GANs are among the most successful generative models (especially in terms of their ability to generate realistic high-resolution images). GANs have been successfully applied to a wide variety of tasks (mostly in research settings) but continue to present unique challenges and research opportunities because they are based on game theory while most other approaches to generative modeling are based on optimization.

1. INTRODUCTION

Most current approaches to developing artificial intelligence are based primarily on machine learning. The most widely used and successful form of machine learning to date is supervised learning. Supervised learning algorithms are given a dataset of pairs of example inputs and example outputs. They learn to associate each input with each output

applications of GANs and identify core research problems related to convergence in games necessary to make GANs a reliable technology.

2. GENERATIVE MODELING

The goal of supervised learning is relatively straightforward to specify, and all supervised learning algorithms have essentially the same goal: learn to accurately associate new input examples with the correct outputs. For instance, an object recognition algorithm may associate a photo of a dog with some kind of DOG category identifier.

Unsupervised learning is a less clearly defined branch of machine learning, with many different unsupervised learning algorithms pursuing many different goals. Broadly speaking, the goal of unsupervised learning is to learn something useful by examining a dataset containing unlabeled input examples. Clustering and dimensionality reduction are common examples of unsupervised learning.

Another approach to unsupervised learning is generative modeling. In generative modeling, training examples $\boldsymbol{x}$ are drawn from an unknown distribution $p_{\text{data}}(\boldsymbol{x})$. The goal of a generative modeling algorithm is to learn a $p_{\text{model}}(\boldsymbol{x})$ that approximates $p_{\text{data}}(\boldsymbol{x})$ as closely as possible.

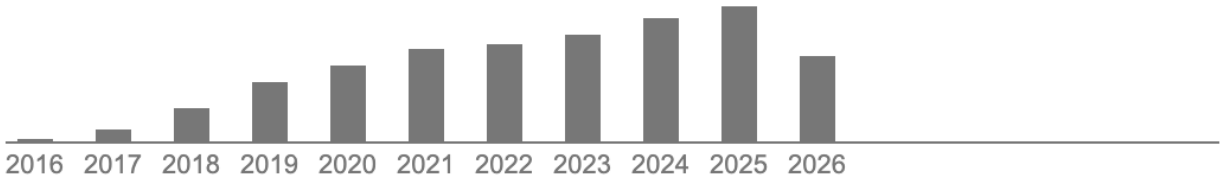
A straightforward way to learn an approximation of p_{data} is to explicitly write a function $p_{\text{model}}(\boldsymbol{x}; \boldsymbol{\theta})$ controlled by parameters $\boldsymbol{\theta}$ and search for the value of the parameters that makes



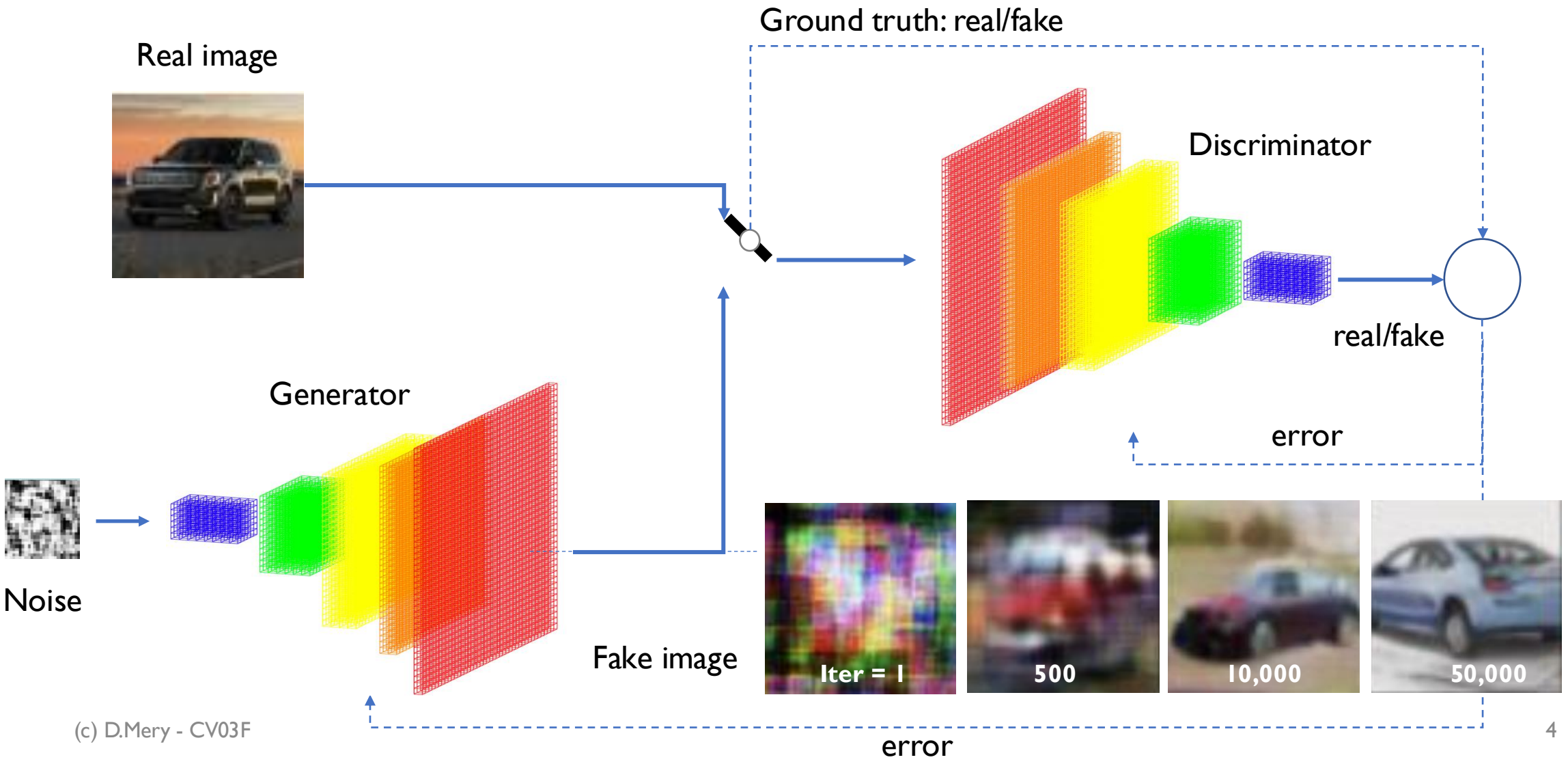
Ian Goodfellow

Generative adversarial networks

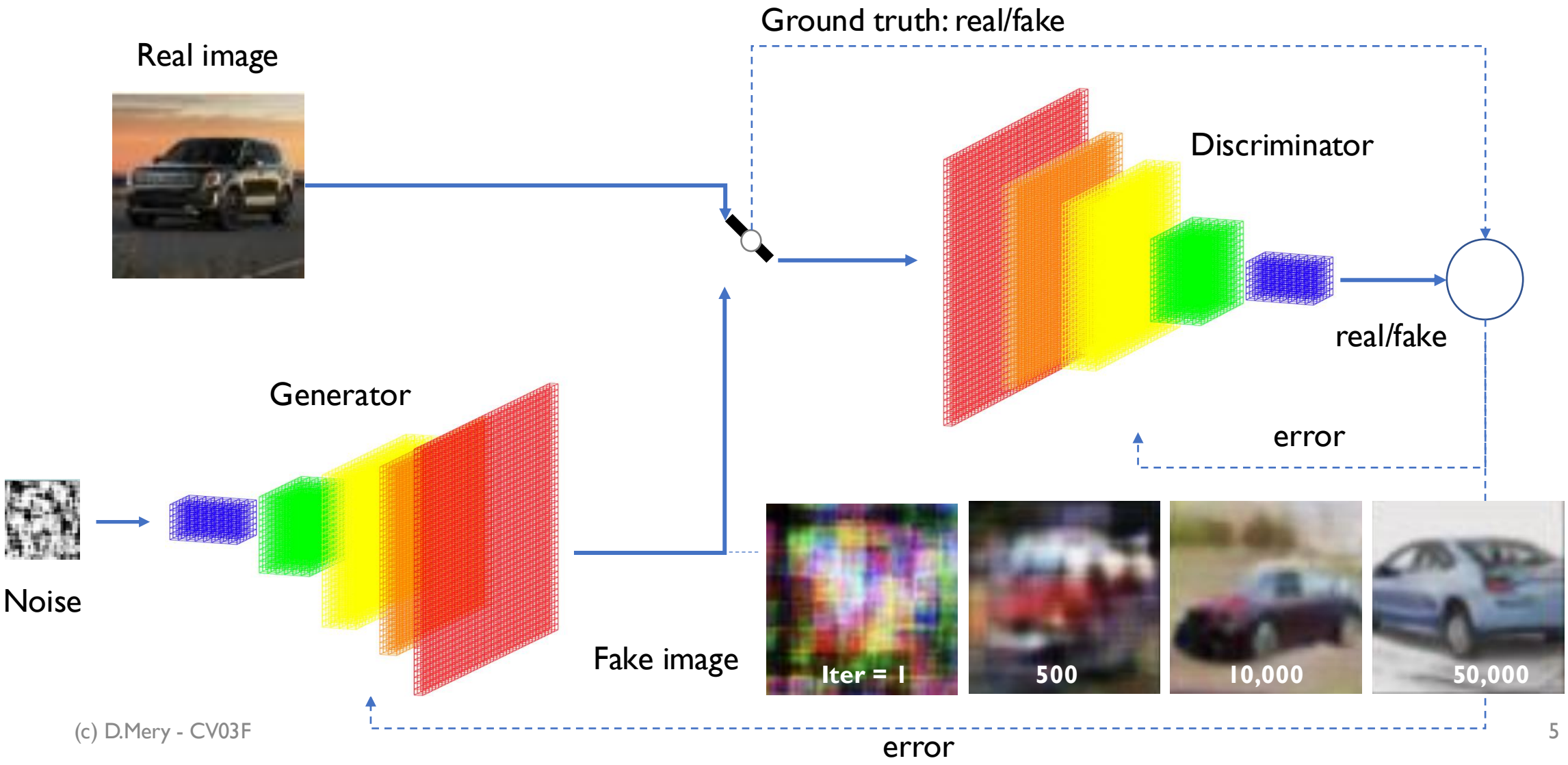
Authors	Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, Yoshua Bengio
Publication date	2014
Conference	Advances in neural information processing systems
Volume	27
Description	We propose a new framework for estimating generative models via adversarial nets, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G. The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D, a unique solution exists, with G recovering the training data distribution and D equal to 1/2 everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitatively evaluation of the generated samples.
Total citations	Cited by 123806

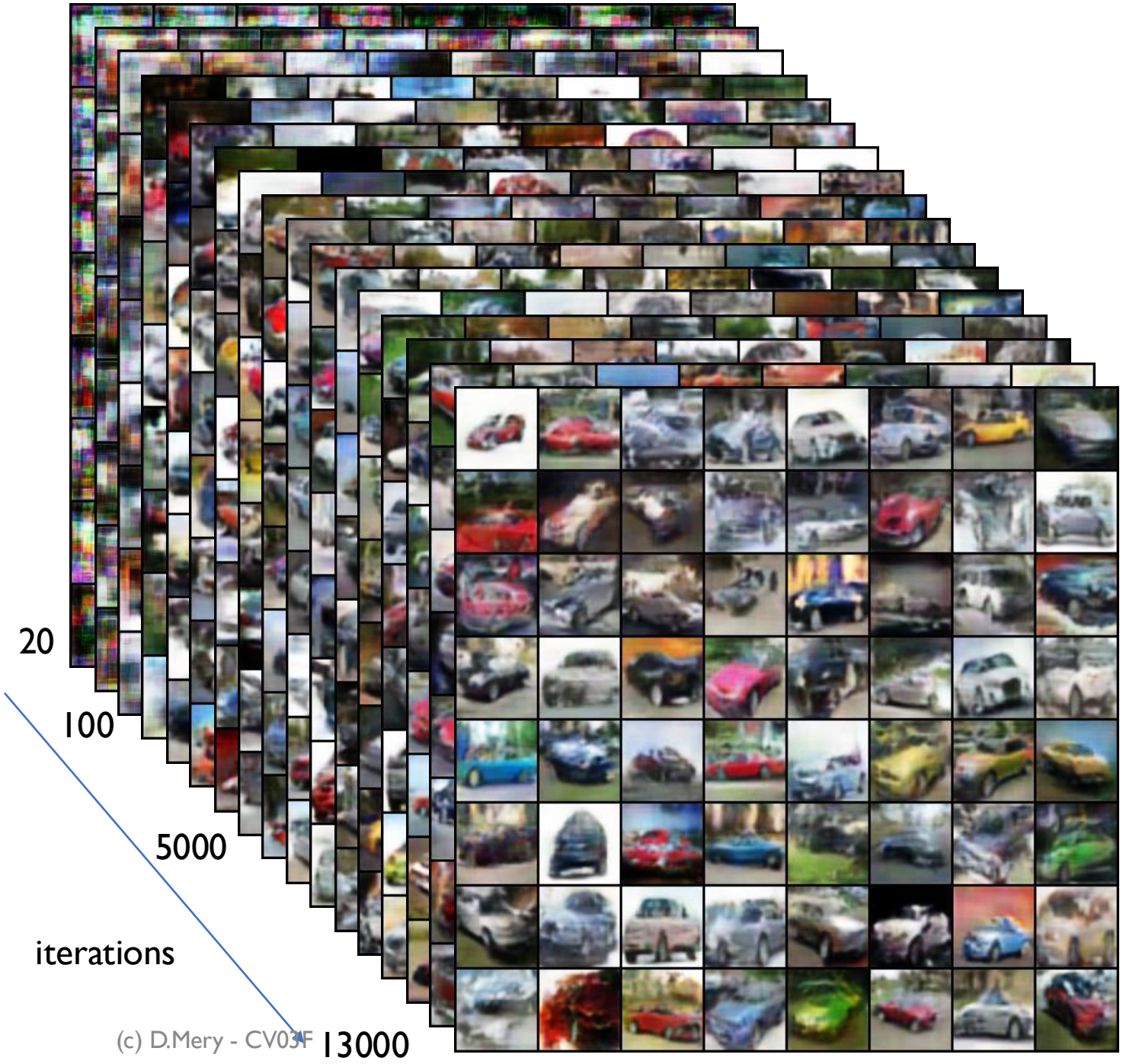


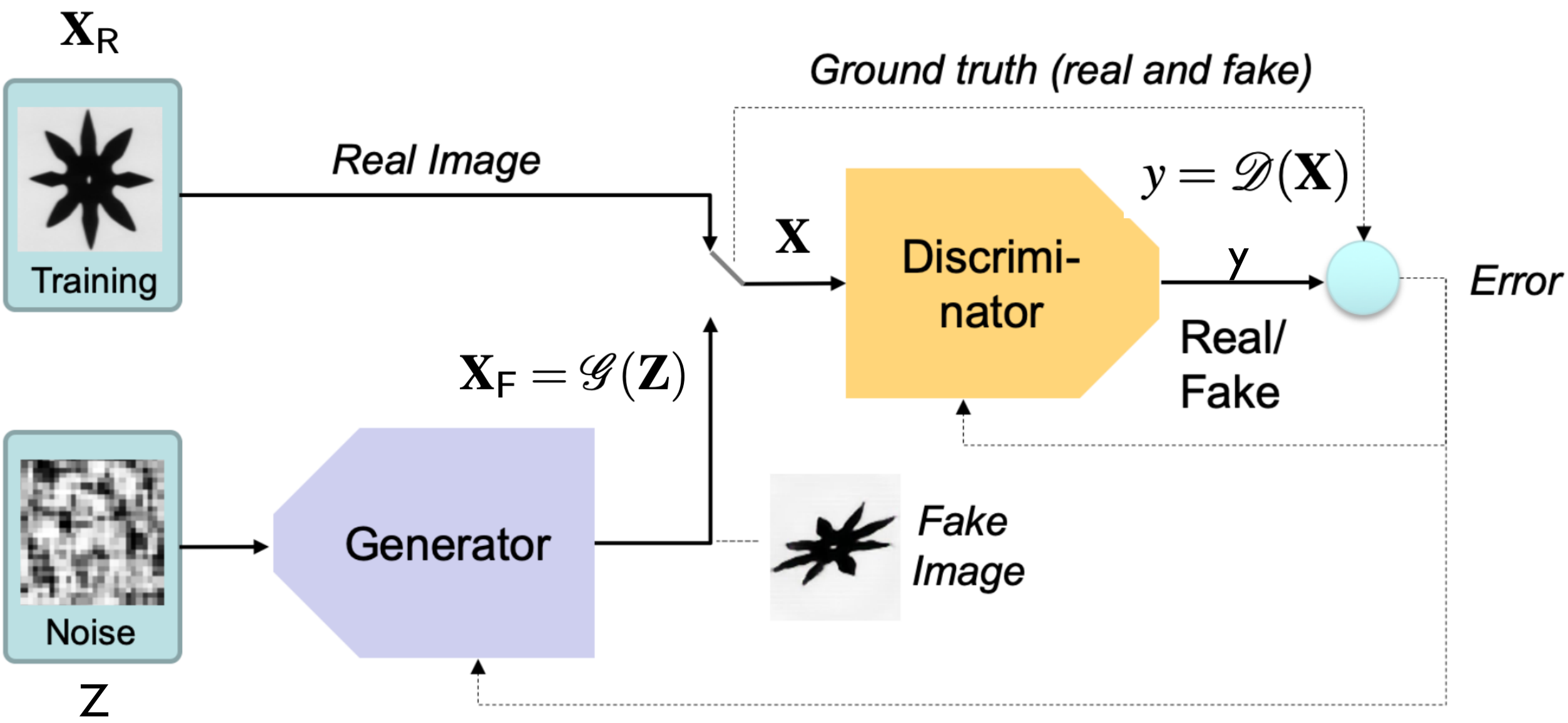
GAN: Generative Adversarial Network

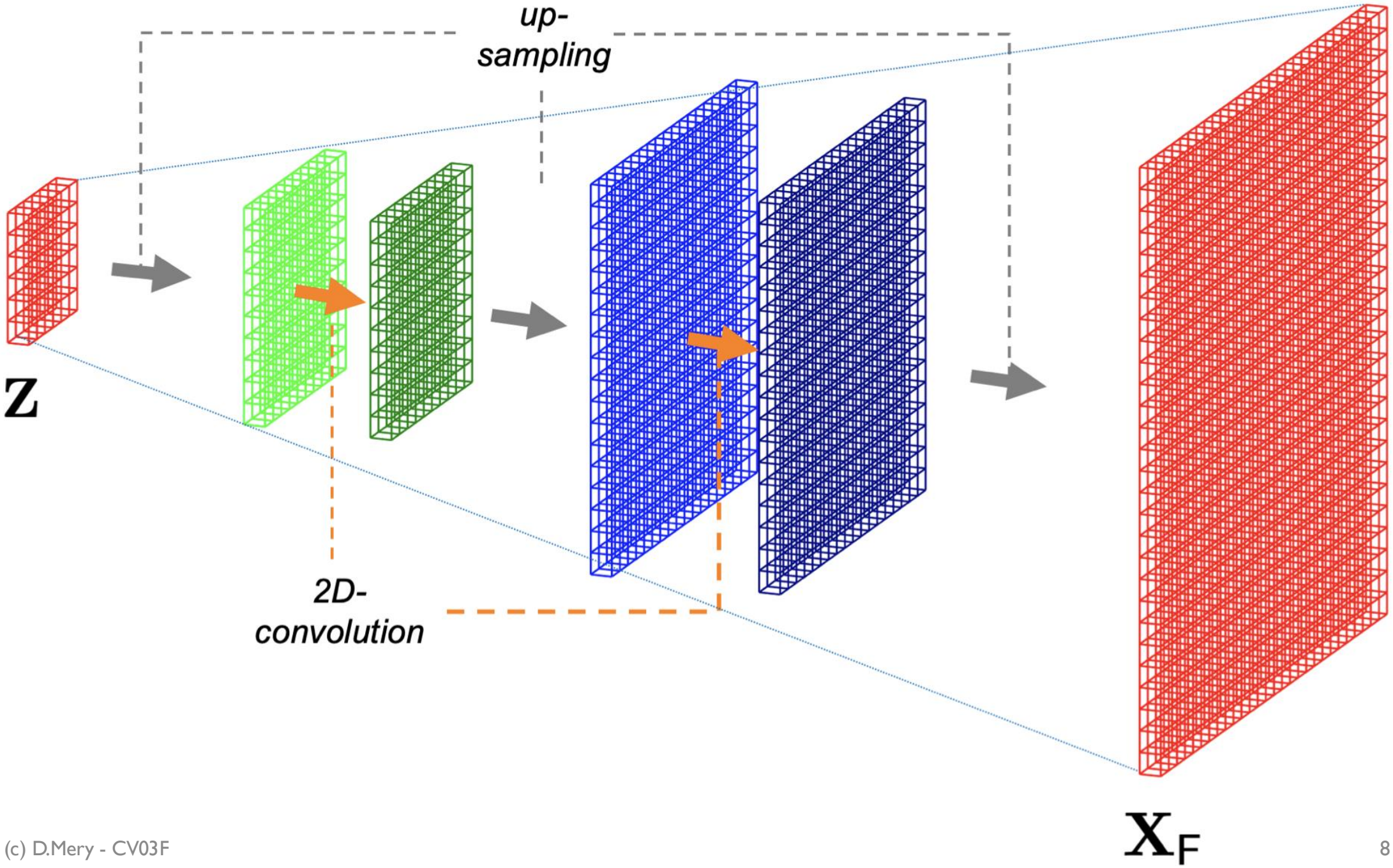


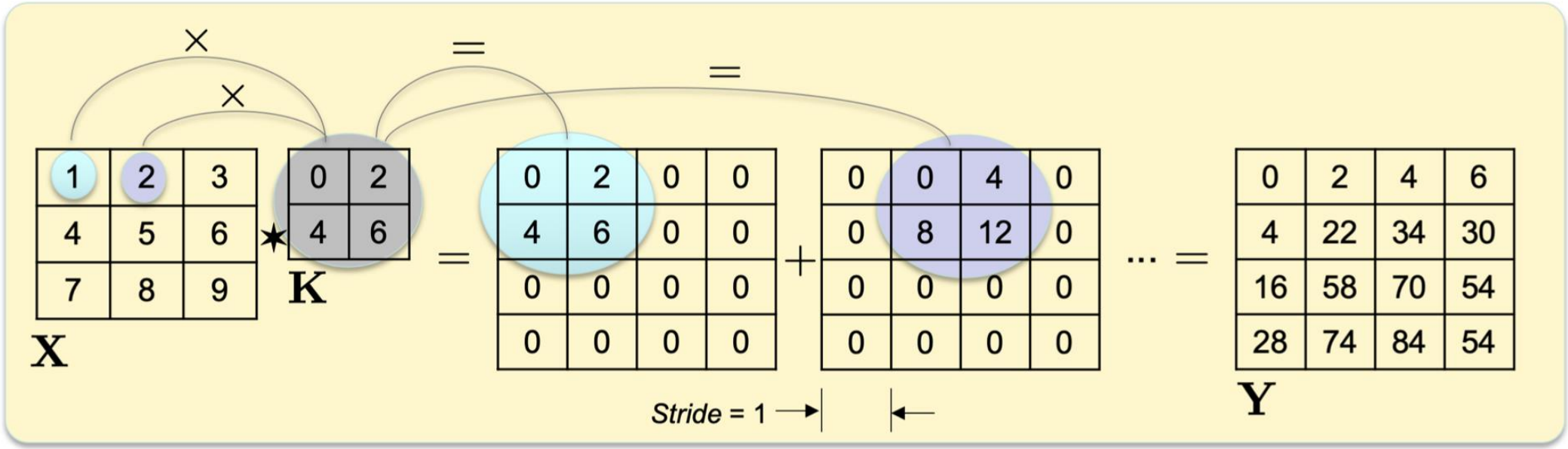
GAN: Generative Adversarial Network

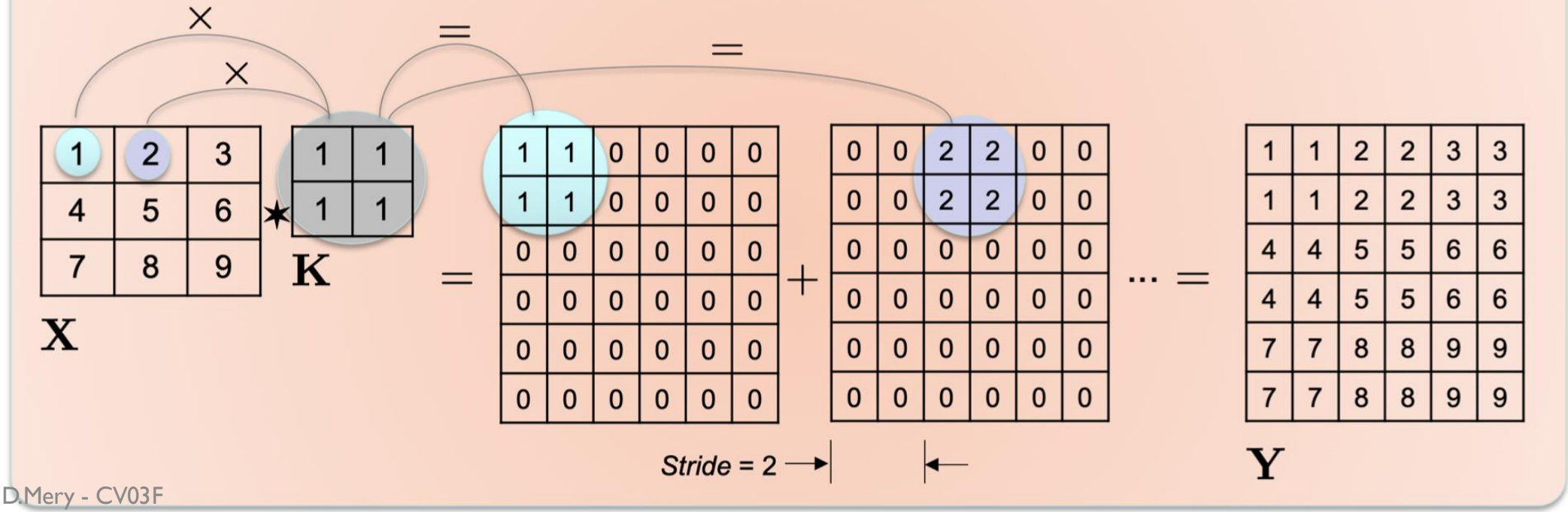


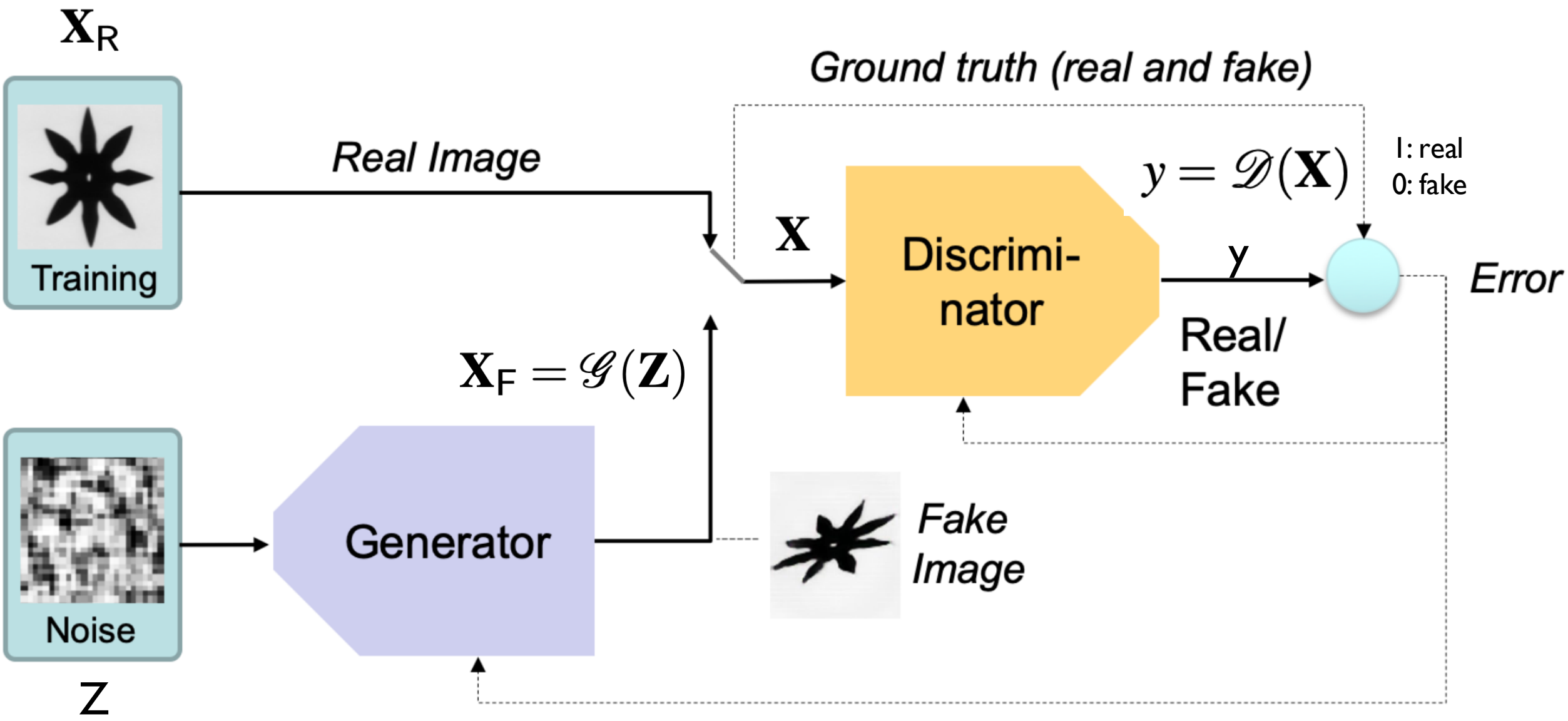

20
100
5000
iterations
13000


$\mathbf{X}_R$
Ground truth (real and fake)
Real Image
$y = \mathscr{D}(\mathbf{X})$
$\mathbf{X}$
Discrimi-
nator
y
Error
Training
Real/
Fake
$\mathbf{X}_F = \mathscr{G}(\mathbf{Z})$
Generator
Fake
Image
Noise
Z


up-
sampling
z
2D-
convolution
X_F


X
K
Stride = 1
Y


X
K
Stride = 2
Y


$\mathbf{X}_R$
Ground truth (real and fake)
Real Image
Training
$y = \mathscr{D}(\mathbf{X})$
1: real
0: fake
$\mathbf{X}$
Discrimi-
nator
y
Error
Real/
Fake
$\mathbf{X}_F = \mathscr{G}(\mathbf{Z})$
Noise
Generator
Fake
Image
Z

In training stage, two sets are available: *i)* a set of m real images, $\mathbf{X}_\mathsf{R}(1)\cdots\mathbf{X}_\mathsf{R}(m)$, and *ii)* a set of m noisy images $\mathbf{Z}(1)\cdots\mathbf{Z}(m)$, from them a set of m fake images are computed $\mathbf{X}_\mathsf{F}(1)\cdots\mathbf{X}_\mathsf{F}(m)$ using (7.42). Thus, the discriminator is learned by maximizing:

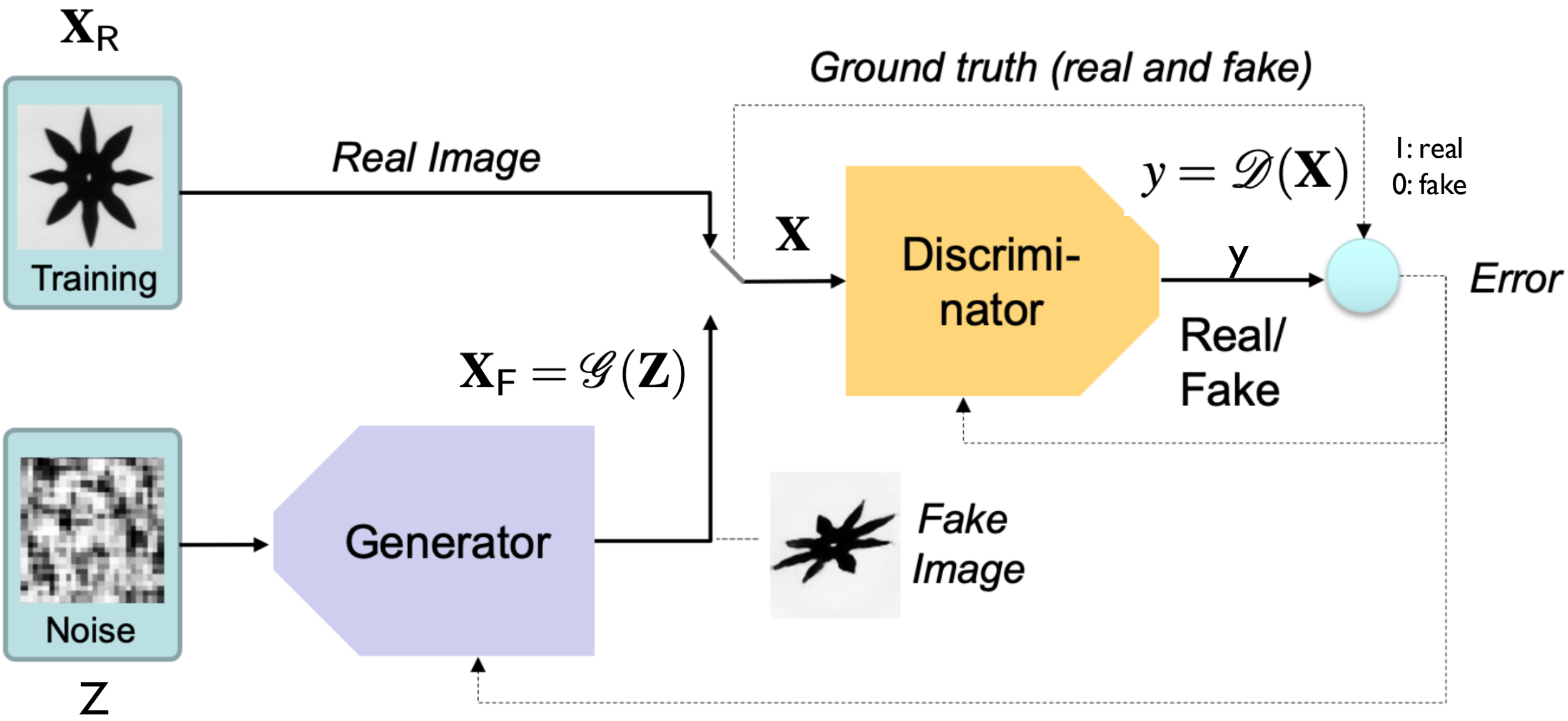
$$J_\mathsf{D} = \sum_{i=1}^{m} \log\left[\mathscr{D}(\mathbf{X}_\mathsf{R}(i)\right] + \sum_{i=1}^{m} \log\left[1 - \mathscr{D}(\mathbf{X}_\mathsf{F}(i)\right] \rightarrow \max \tag{7.44}$$

The first sum is maximal when the real images are classified as 'real', whereas the second sum is maximal when the synthetic images are classified as 'fake'. Thus, the idea of (7.44) is to classify as 1 the real images and as 0 the synthetic ones.

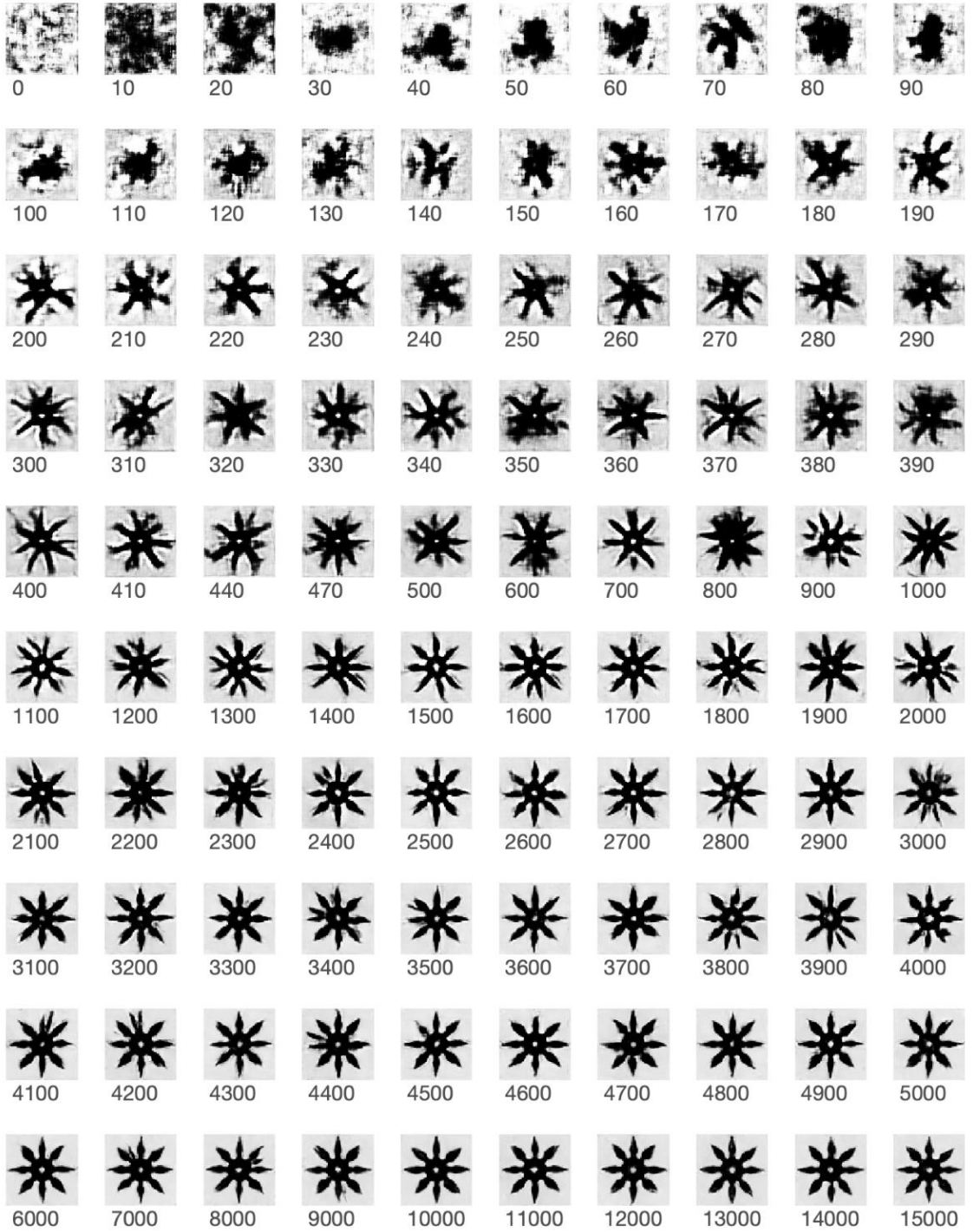
Thus, the generator is trained by minimizing the following objective function:

$$J_{\mathsf{G}} = \sum_{i=1}^{m} \log\left[1 - \mathscr{D}(\mathbf{X}_{\mathsf{F}}(i)\right] \rightarrow \min \tag{7.45}$$

In this case, the sum is minimal when the synthetic images are classified as 'real'. That means that the generated fake images will be so realistic that the discriminator will classified them as 'real'. In the training stage, objective functions J_{D} and J_{G} are playing a *min-max* game, the reader can see that the second sum of (7.44) is equal to the sum of (7.45), however, in first case we are trying to maximize it (the discriminator should recognize that the synthetic images are fake), whereas in the second one the aim is to minimize it (the generator wants to fool the discriminator).


$\mathbf{X}_R$
Ground truth (real and fake)
Real Image
Training
$y = \mathscr{D}(\mathbf{X})$
1: real
0: fake
$\mathbf{X}$
Discrimi-
nator
y
Error
Real/
Fake
$\mathbf{X}_F = \mathscr{G}(\mathbf{Z})$
Noise
Generator
Fake
Image
$\mathbf{Z}$

GAN: Shuriken



GAN: MINST



www.ThisPearsonDoesNotExist.com









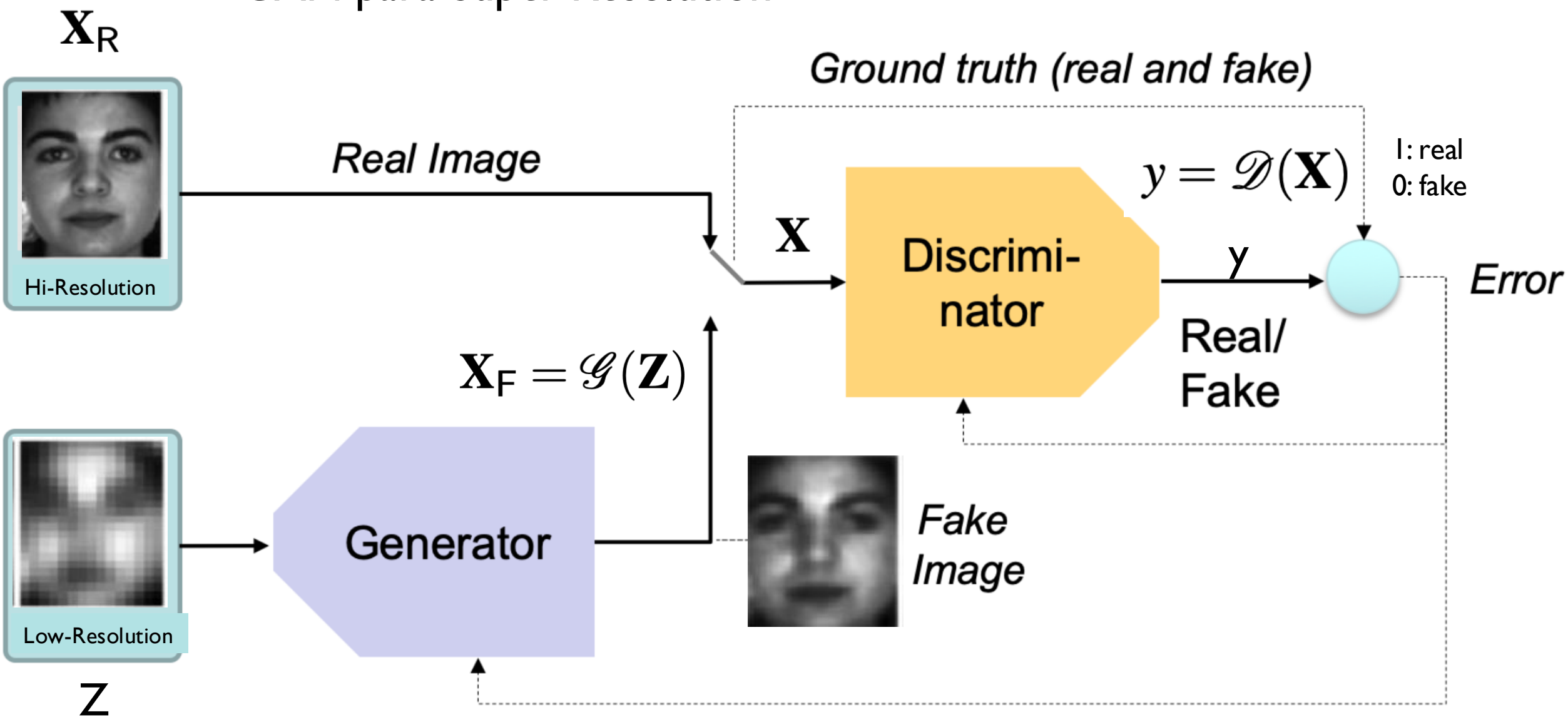








GAN para Super-Resolution



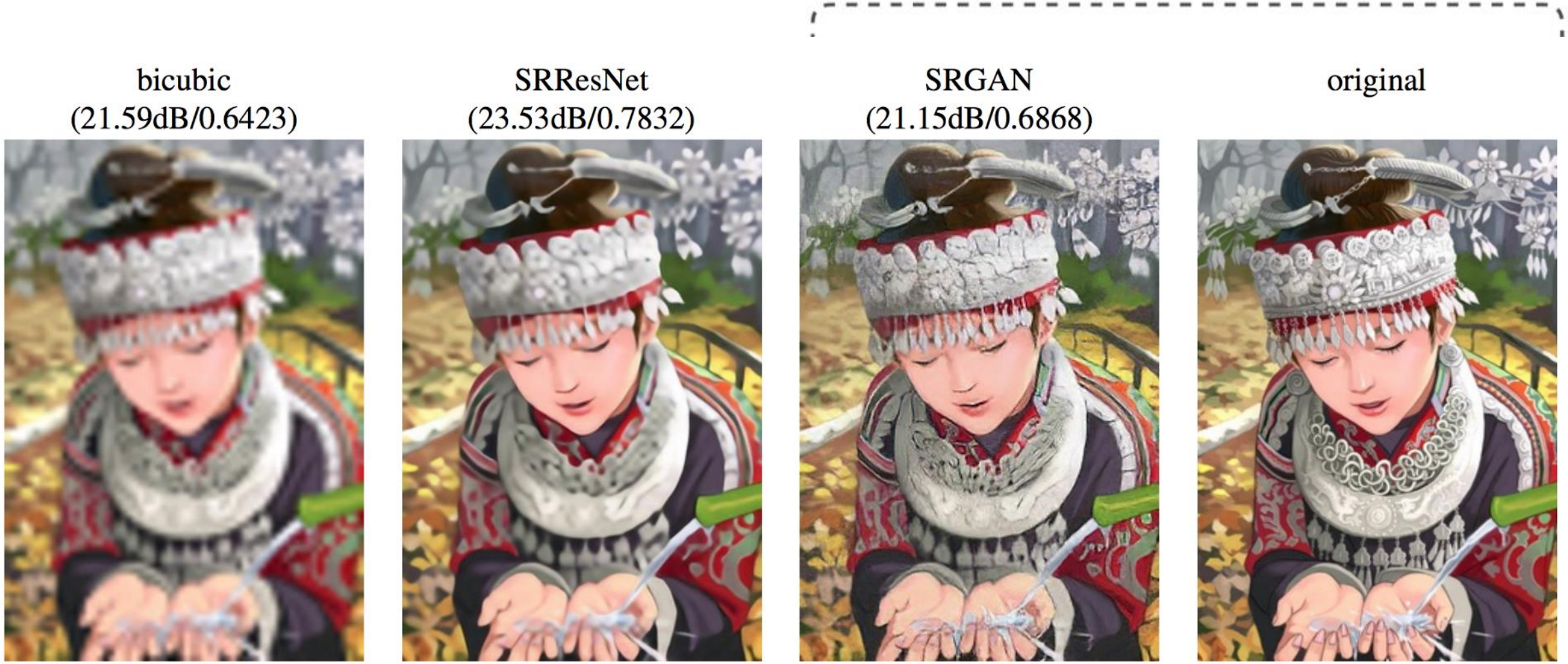
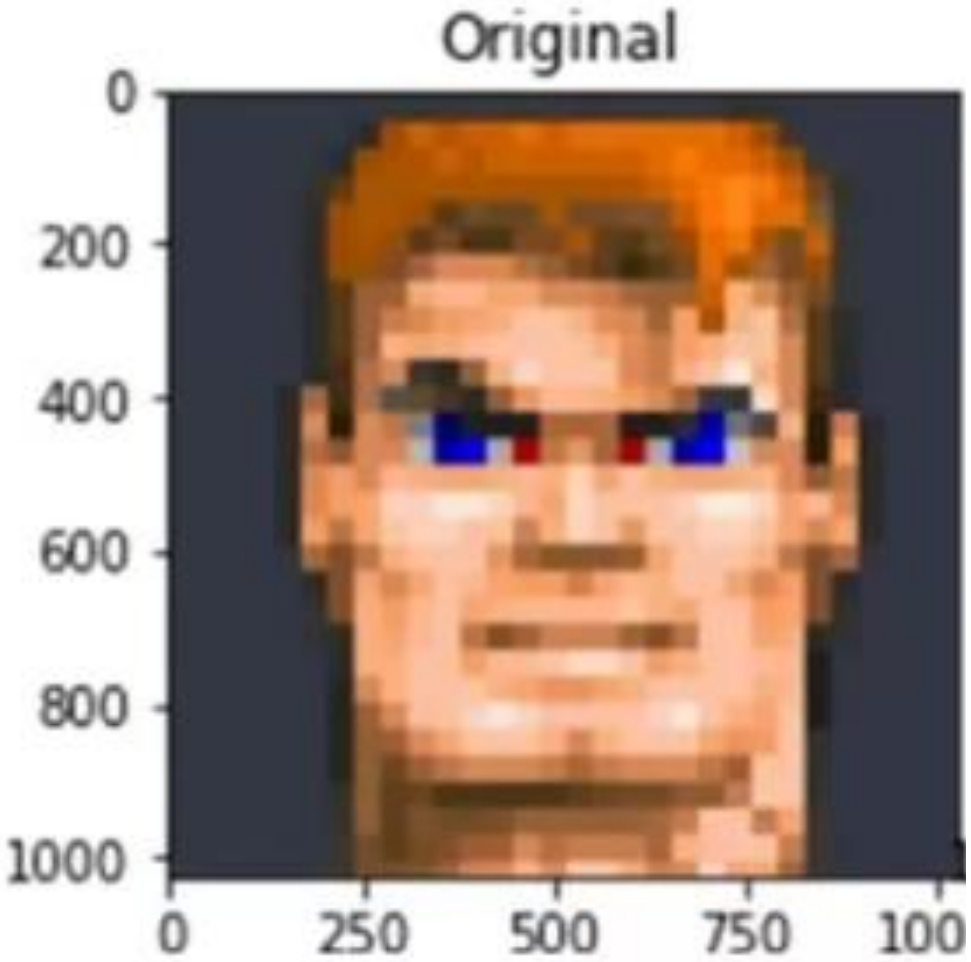
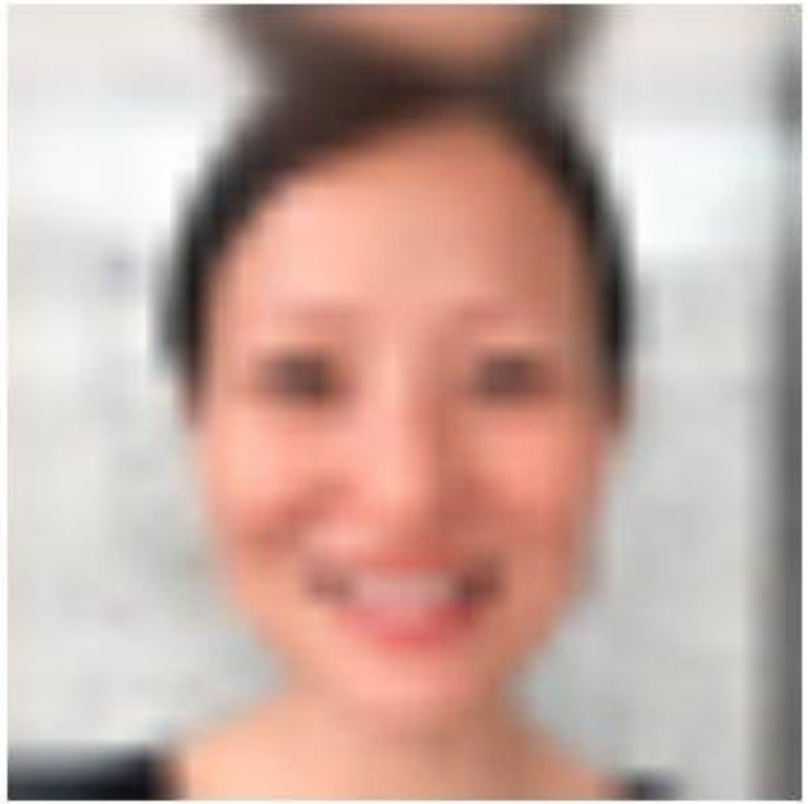


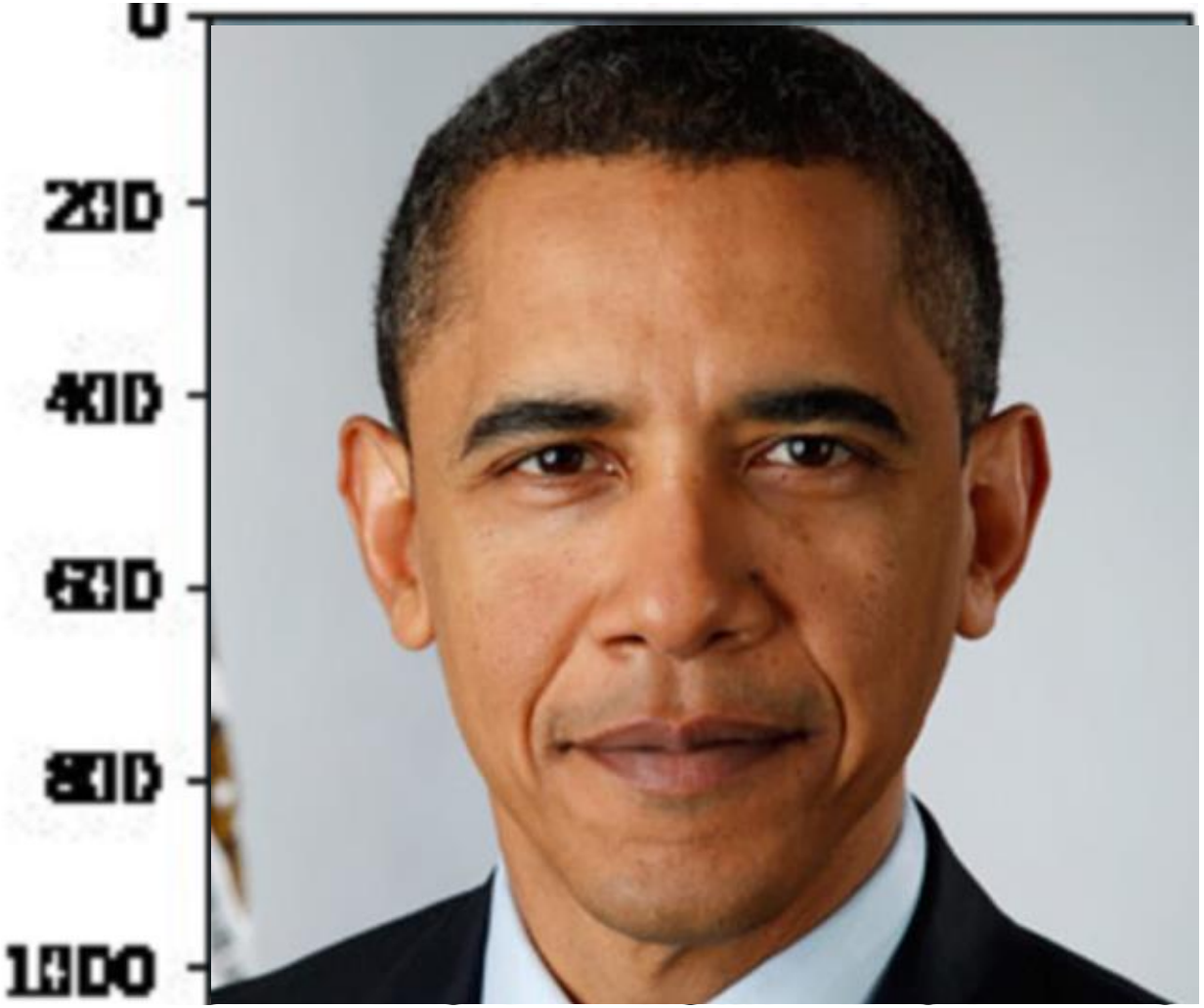
Figure 2: From left to right: bicubic interpolation, deep residual network optimized for MSE, deep residual generative adversarial network optimized for a loss more sensitive to human perception, original HR image. Corresponding PSNR and SSIM are shown in brackets. [4× upscaling]




Original
0
200
400
600
800
1000
0
250
500
750
100

Algunos problemitas...





GAN for Super-Resolution



ORIGINAL
448x448



INPUT
56x56



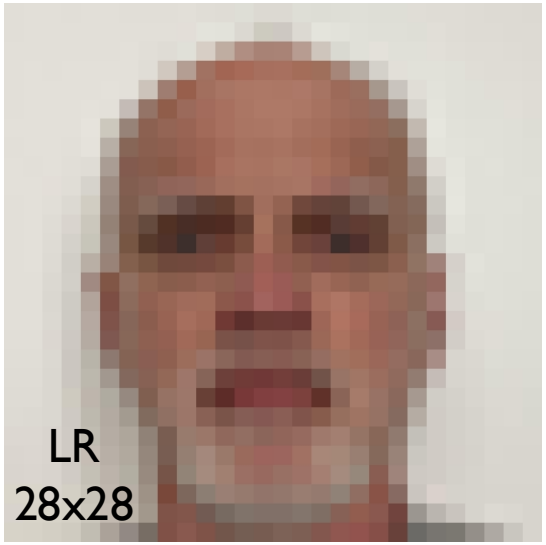
OUTPUT
448x448

Problems with GANs for Super-Resolution

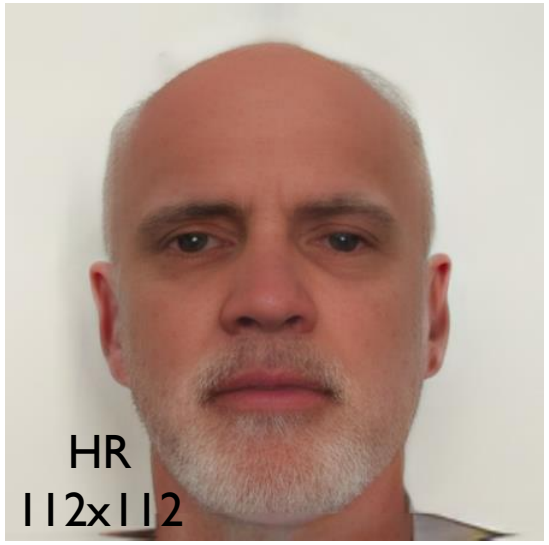


ORIGINAL
112x112





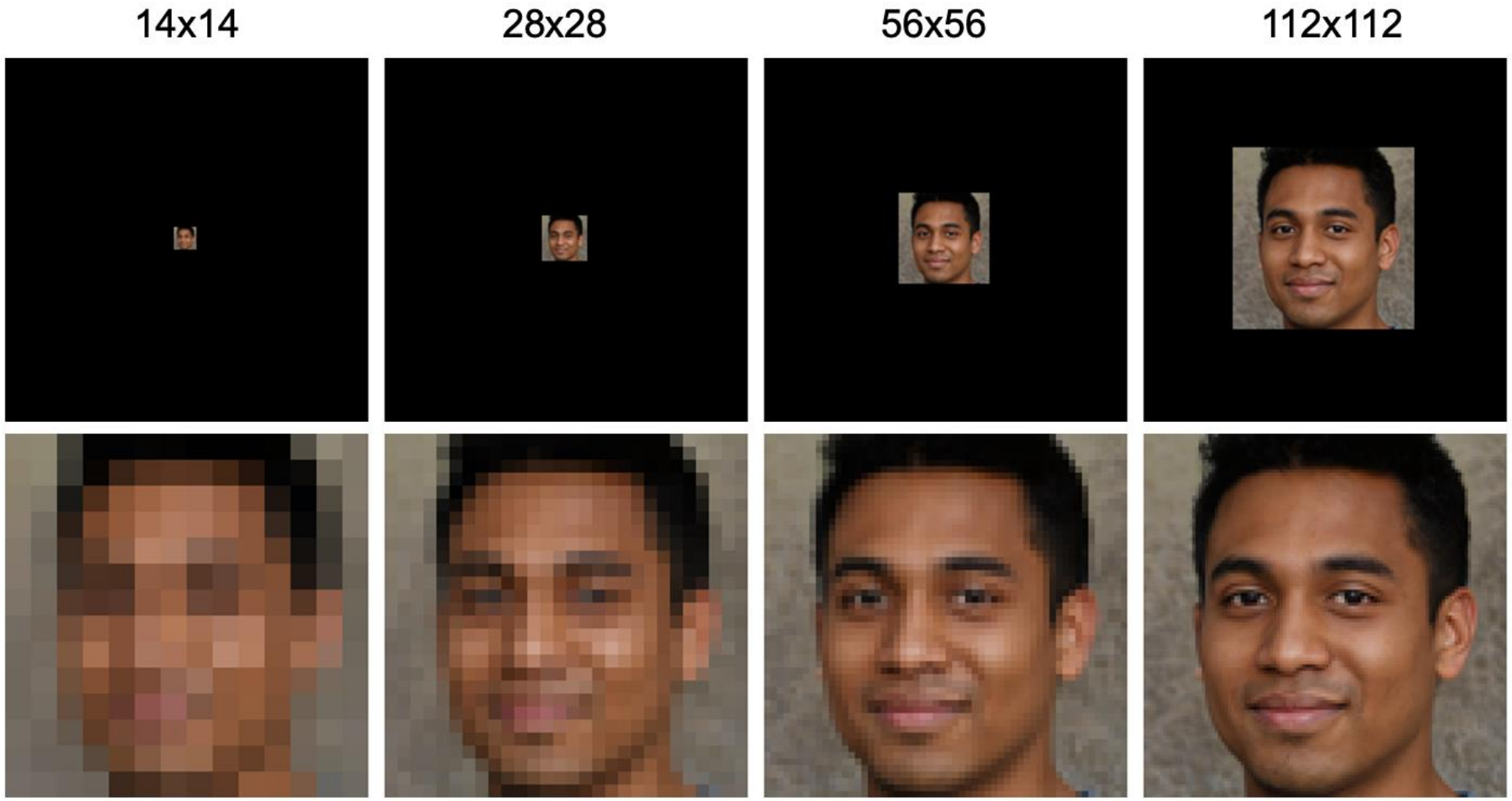




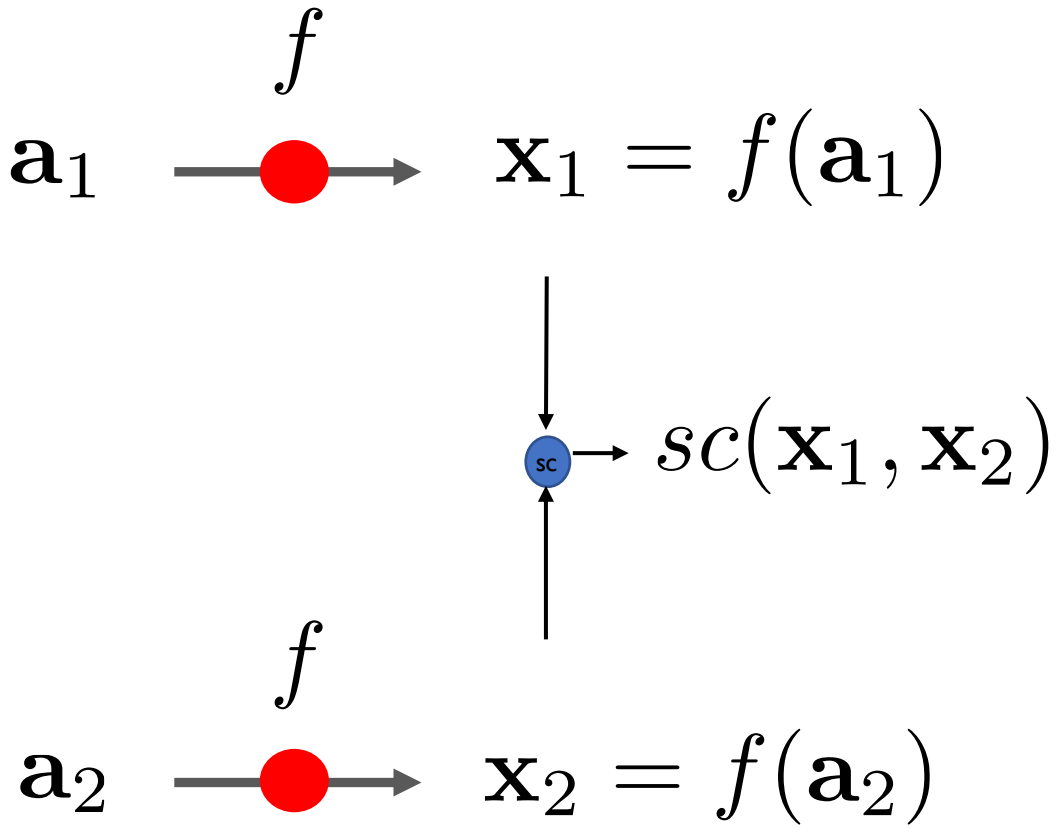




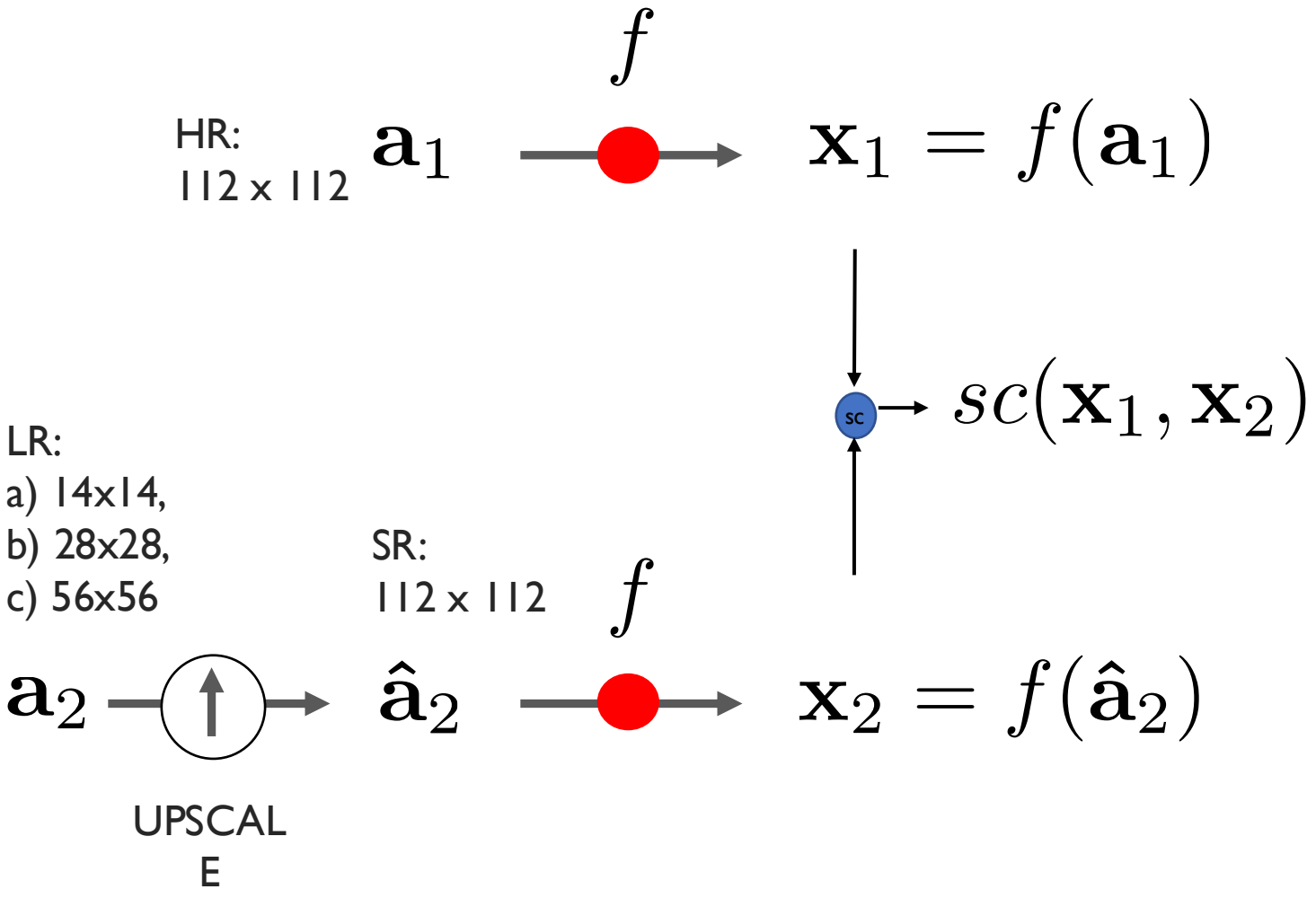
Example of face images with different sizes



Standard face verification approach



Low-resolution face verification approach

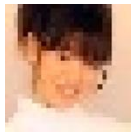


Prieto, L., Pulgar, S., Flynn, P., & Mery, D. (2022, November). On Low-Resolution Face Re-identification with High-Resolution-Mapping. In Pacific-Rim Symposium on Image and Video Technology (PSIVT2022). (pp. 89-102). Cham: Springer International Publishing.

Super-resolution examples 28x28

input

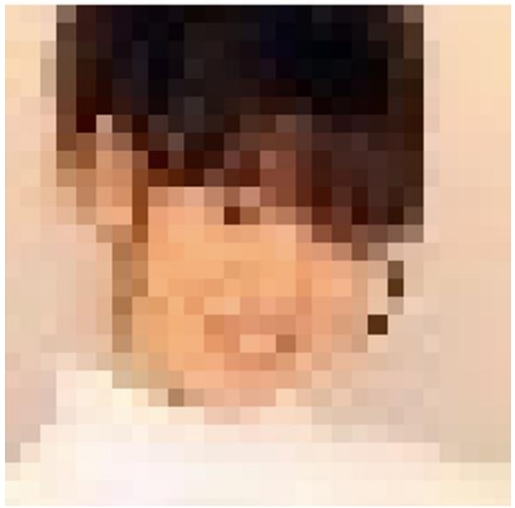
28x28



Super-resolution examples 28x28

input

28x28



Super-resolution examples 28x28



Super-resolution examples 14x14

input

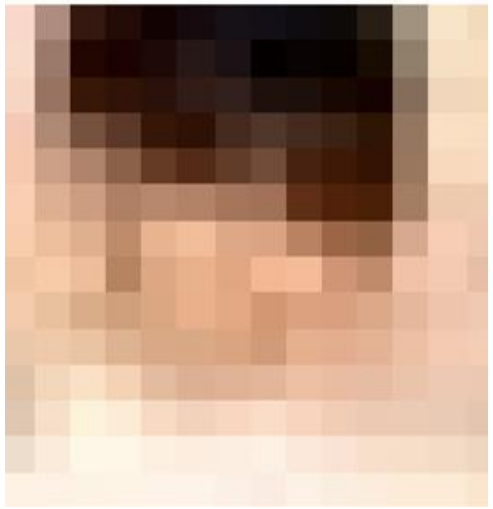
14x14



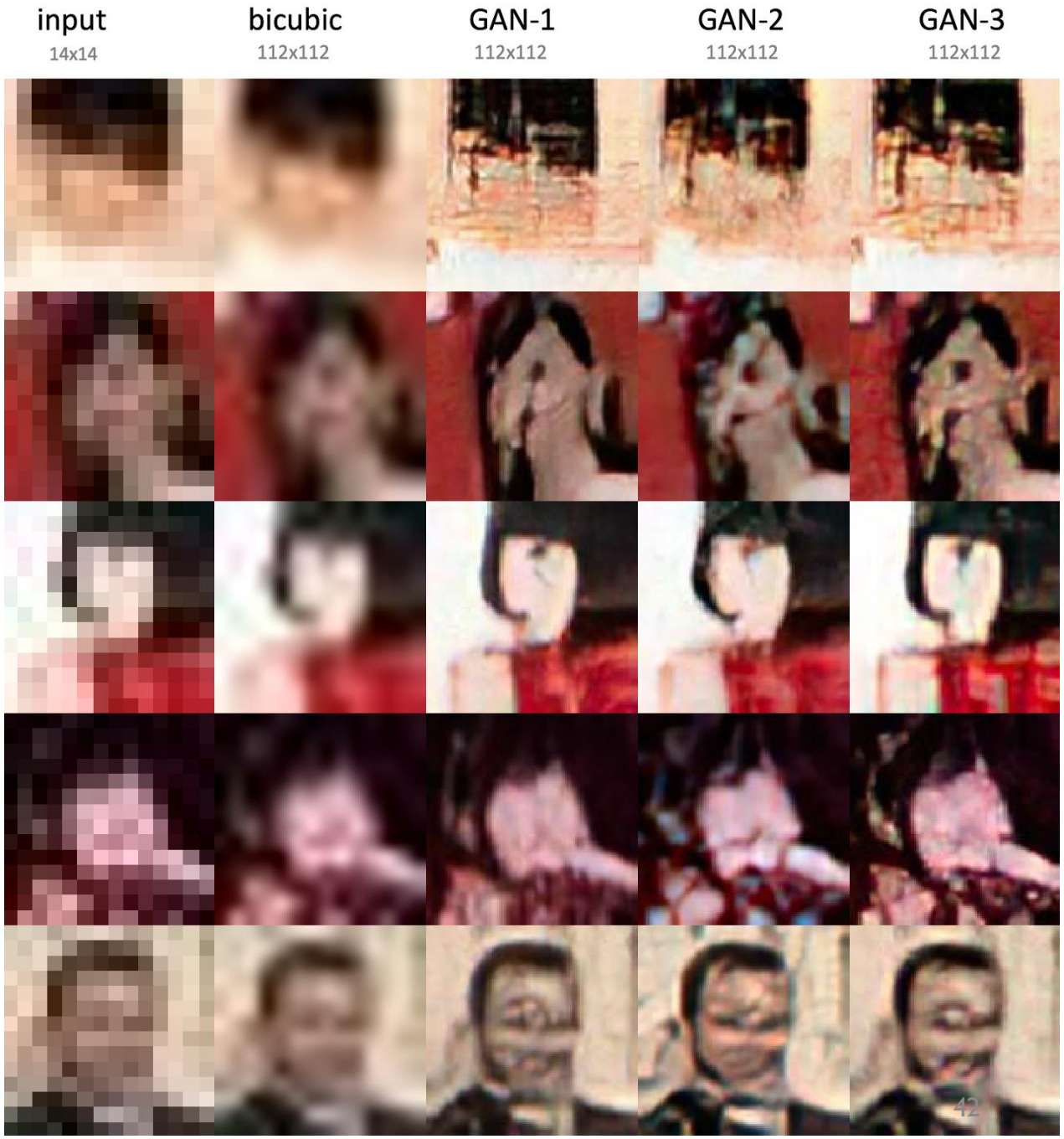
Super-resolution examples 14x14

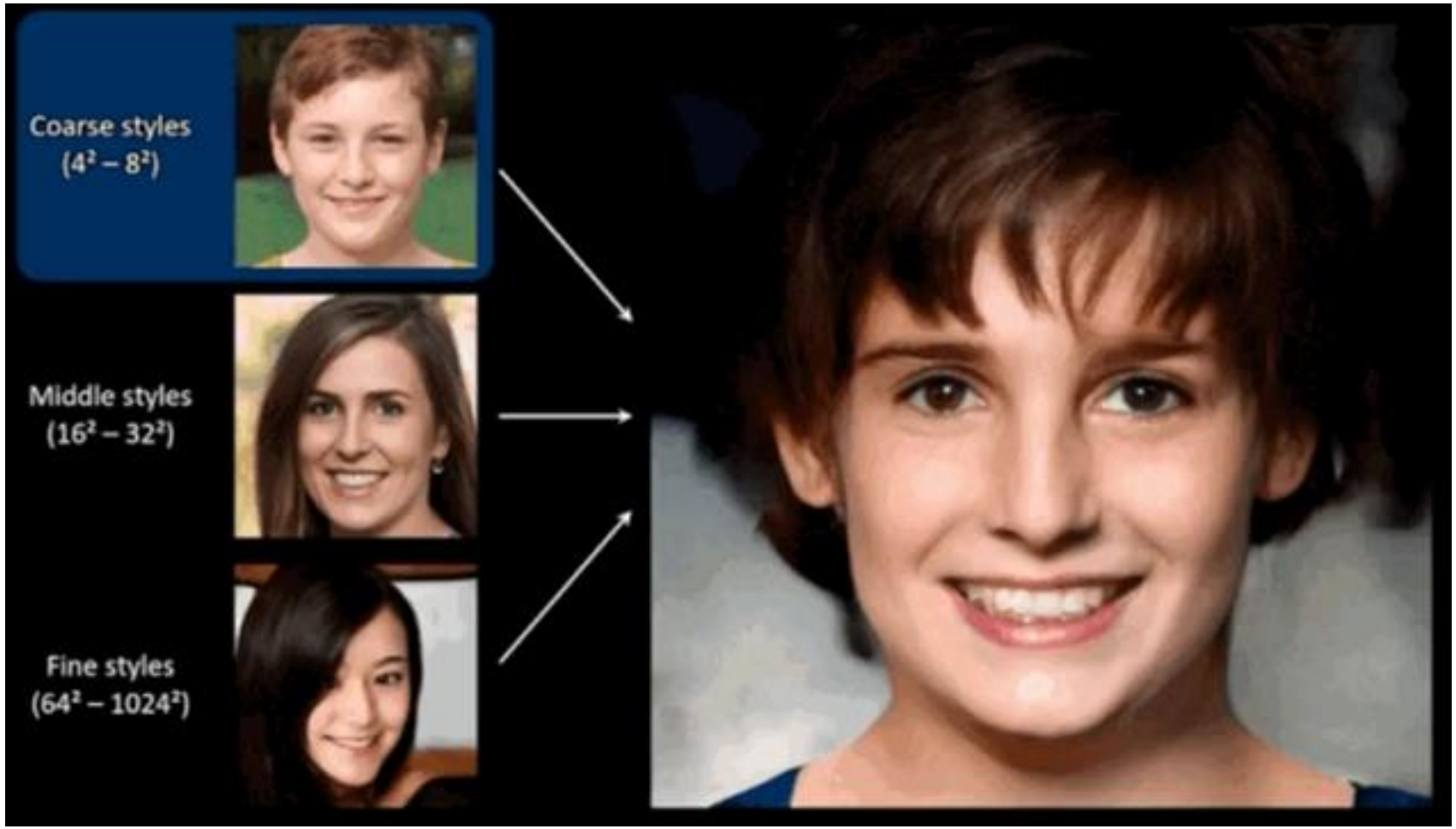
input

14x14

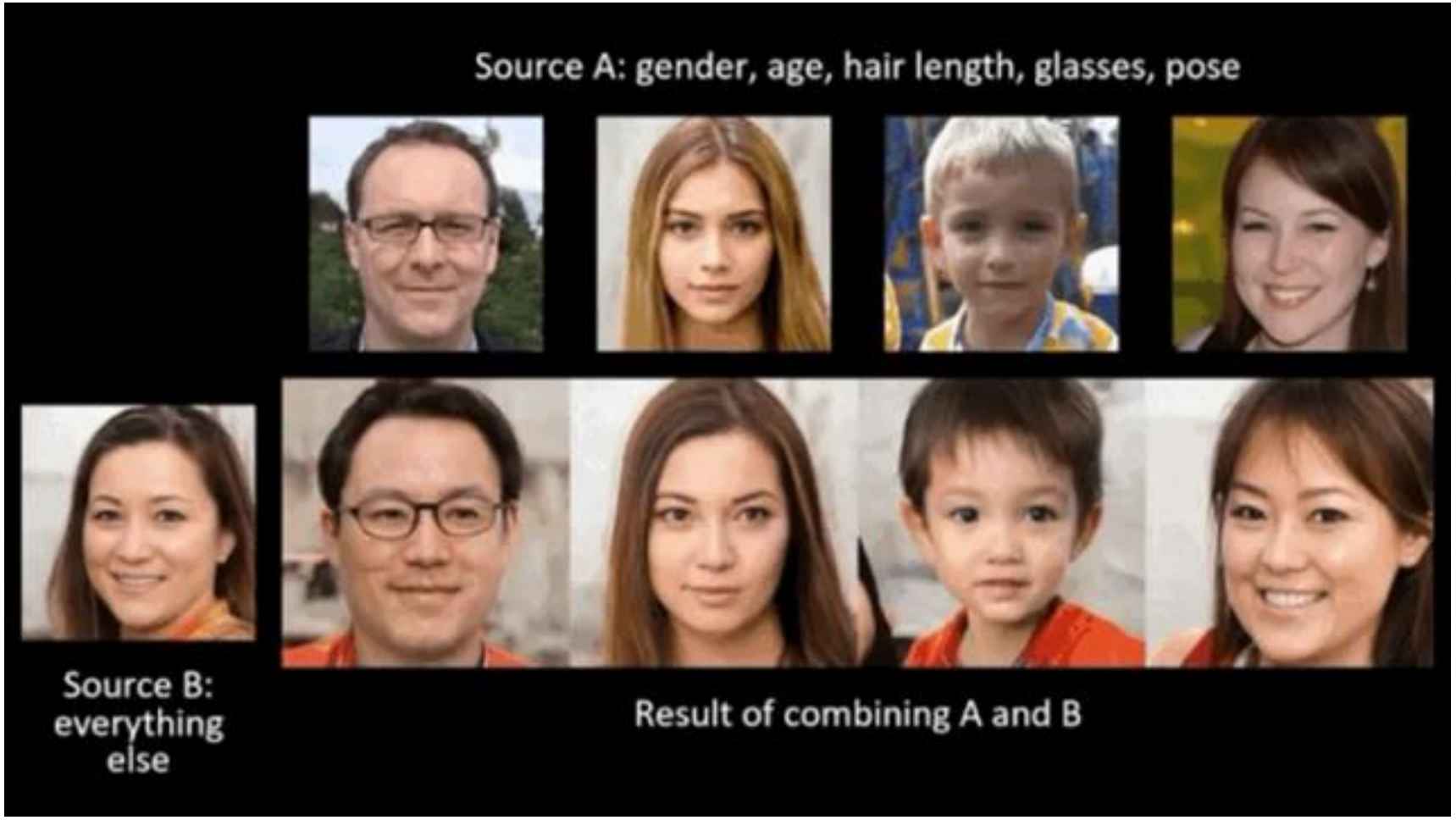


Super-resolution examples 14x14





https://syncedreview.com/2019/02/09/nvidia-open-sources-hyper-realistic-face-generator-stylegan/



https://syncedreview.com/2019/02/09/nvidia-open-sources-hyper-realistic-face-generator-stylegan/

8. (



FakeApp: ¿cómo hacer un DeepFake y cómo funciona la app? - Cultura Geek



8 steps to verify Deep Fake videos | by Konrad Weber | Medium


Powered by Synthesia

Detección de Anomalías

f-AnoGAN: Fast unsupervised anomaly detection with generative adversarial networks

Thomas Schlegl[a,b], Philipp Seeböck[a,b], Sebastian M. Waldstein[b], Georg Langs[a,*], Ursula Schmidt-Erfurth[b]

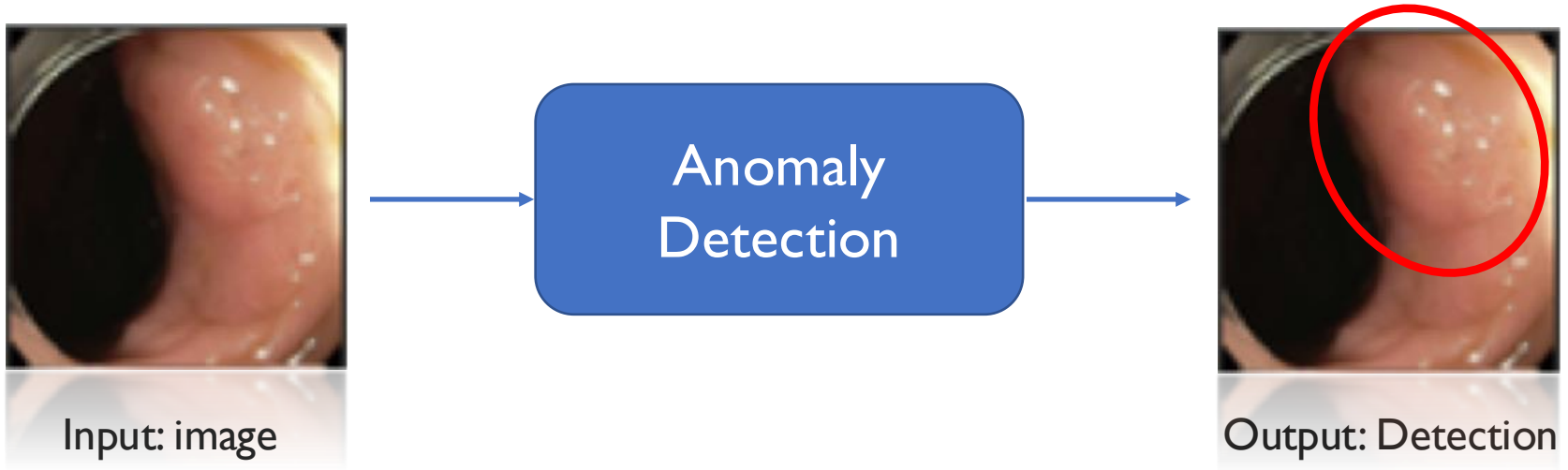
[a] *Computational Imaging Research Lab, Department of Biomedical Imaging and Image-guided Therapy, Medical University of Vienna, Vienna, Austria*
[b] *Christian Doppler Laboratory for Ophthalmic Image Analysis, Department of Ophthalmology and Optometry, Medical University Vienna, Austria*

Abstract

Obtaining expert labels in clinical imaging is difficult since exhaustive annotation is time-consuming. Furthermore, not all possibly relevant markers may be known and sufficiently well described a priori to even guide annotation. While supervised learning yields good results if expert labeled training data is available, the visual variability, and thus the vocabulary of findings, we can detect and exploit, is limited to the annotated lesions. Here, we present *fast AnoGAN* (*f-AnoGAN*), a generative adversarial network (GAN) based unsupervised learning approach capable of identifying anomalous images and image segments, that can serve as imaging biomarker candidates. We build a generative model of healthy training data, and propose and evaluate a fast mapping technique of new data to the GAN's latent space. The mapping is based on a trained encoder, and anomalies are detected via a combined anomaly score based on the building blocks of the trained model – comprising a discriminator feature residual error and an image reconstruction error. In the experiments on optical coherence tomography data, we compare the proposed method with alternative approaches, and provide comprehensive empirical evidence that *f-AnoGAN* outperforms alternative approaches and yields high anomaly detection accuracy. In addition, a visual Turing test with two retina experts showed that the generated images are indistinguishable from real normal retinal OCT images. The f-AnoGAN code is available at https://github.com/tSchlegl/f-AnoGAN.

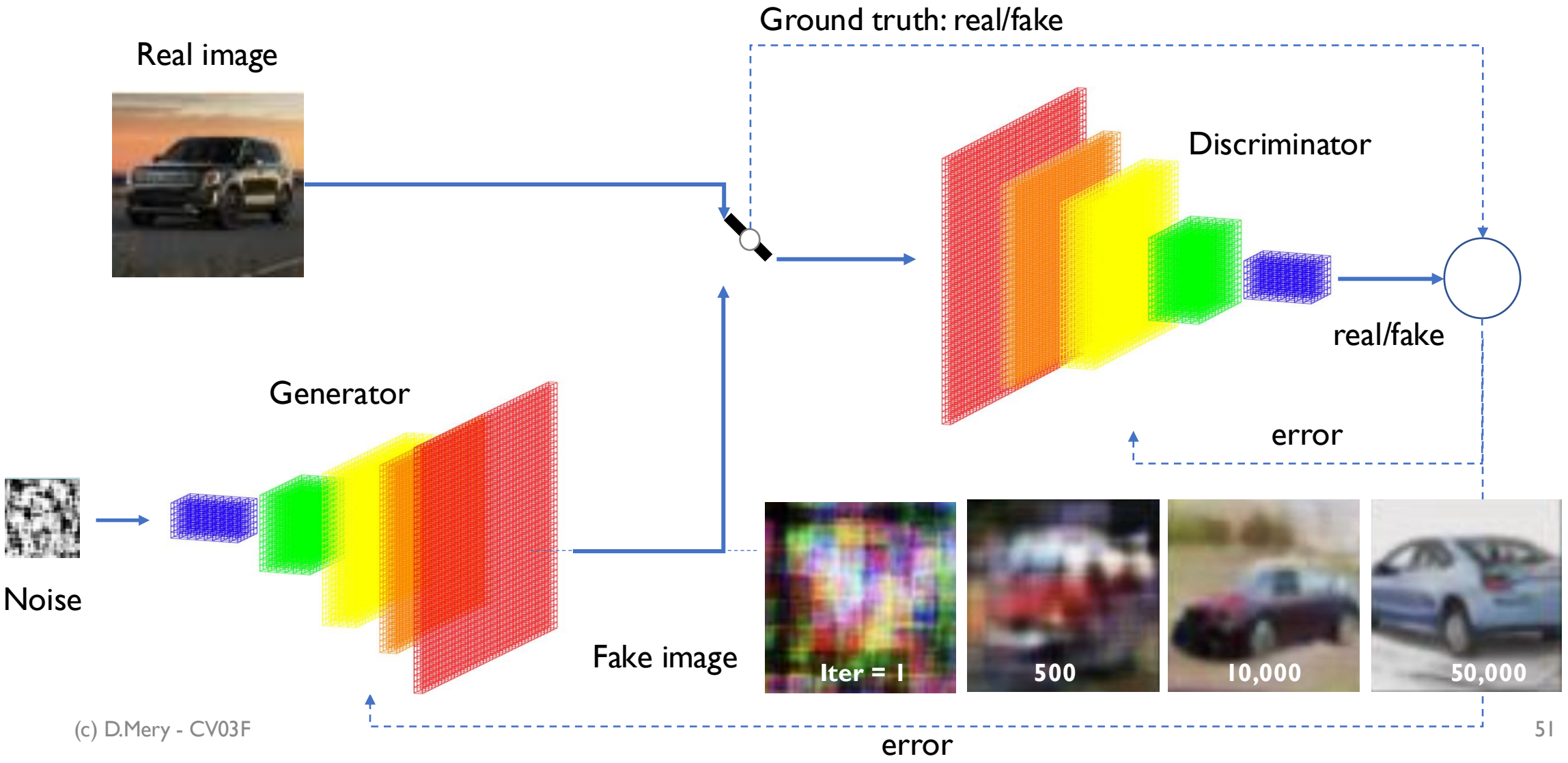
Anomaly Detection – Problem

> Detect an anomaly in an image



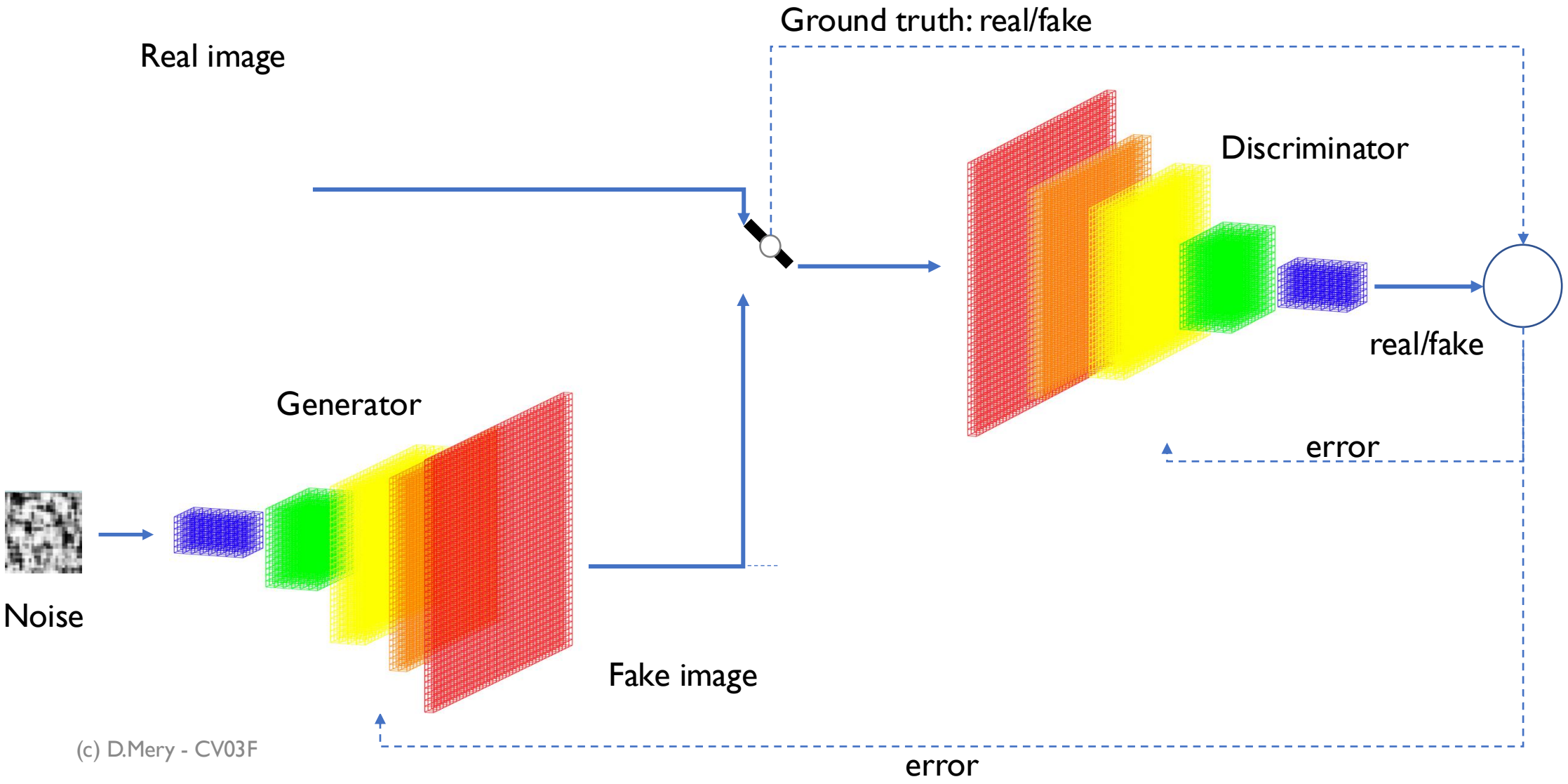
Anomaly Detection – Solution

Deep Learning Model: GAN
Generative Adversarial Network



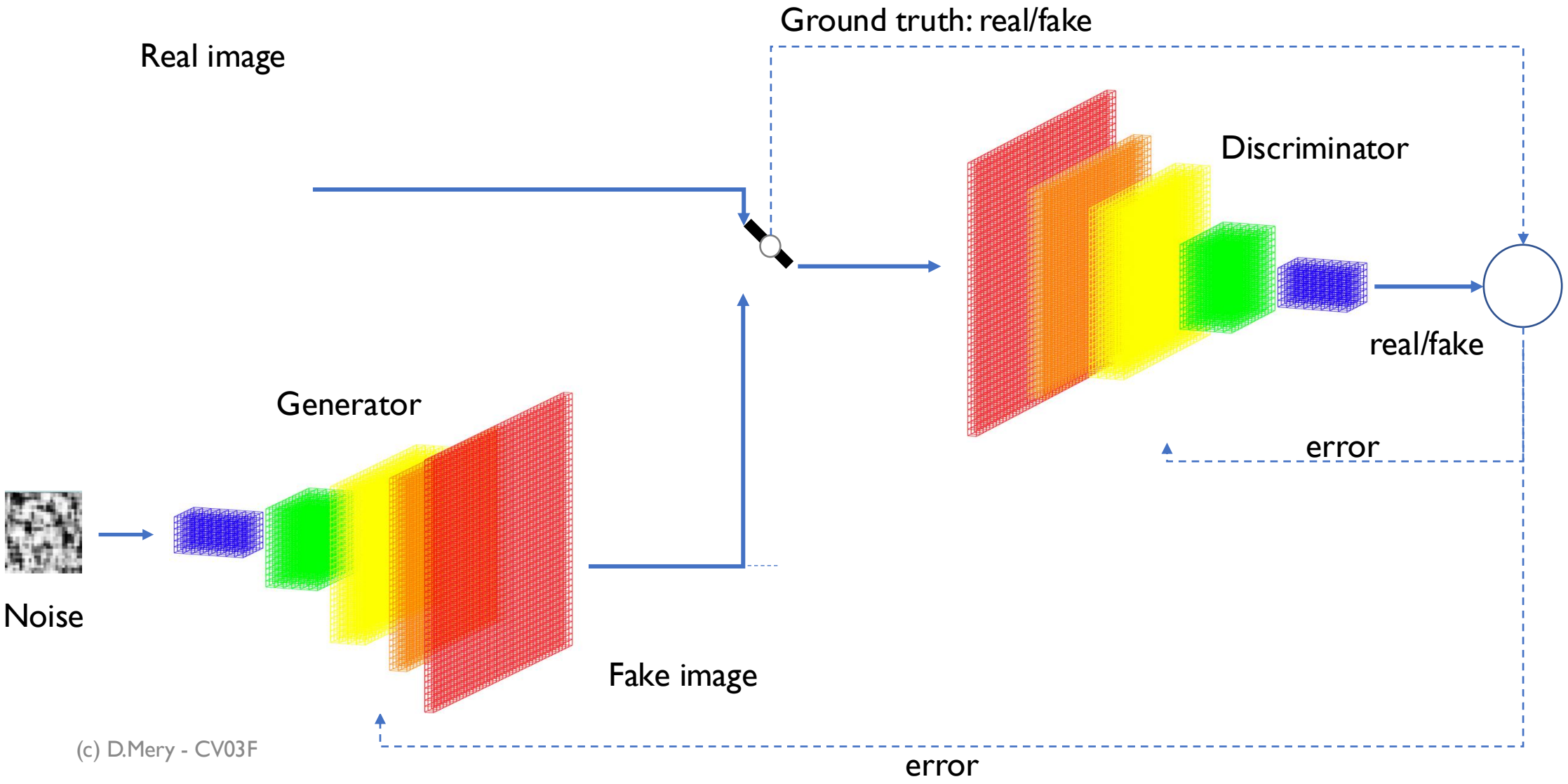
Anomaly Detection – Solution

Deep Learning Model: GAN
Generative Adversarial Network



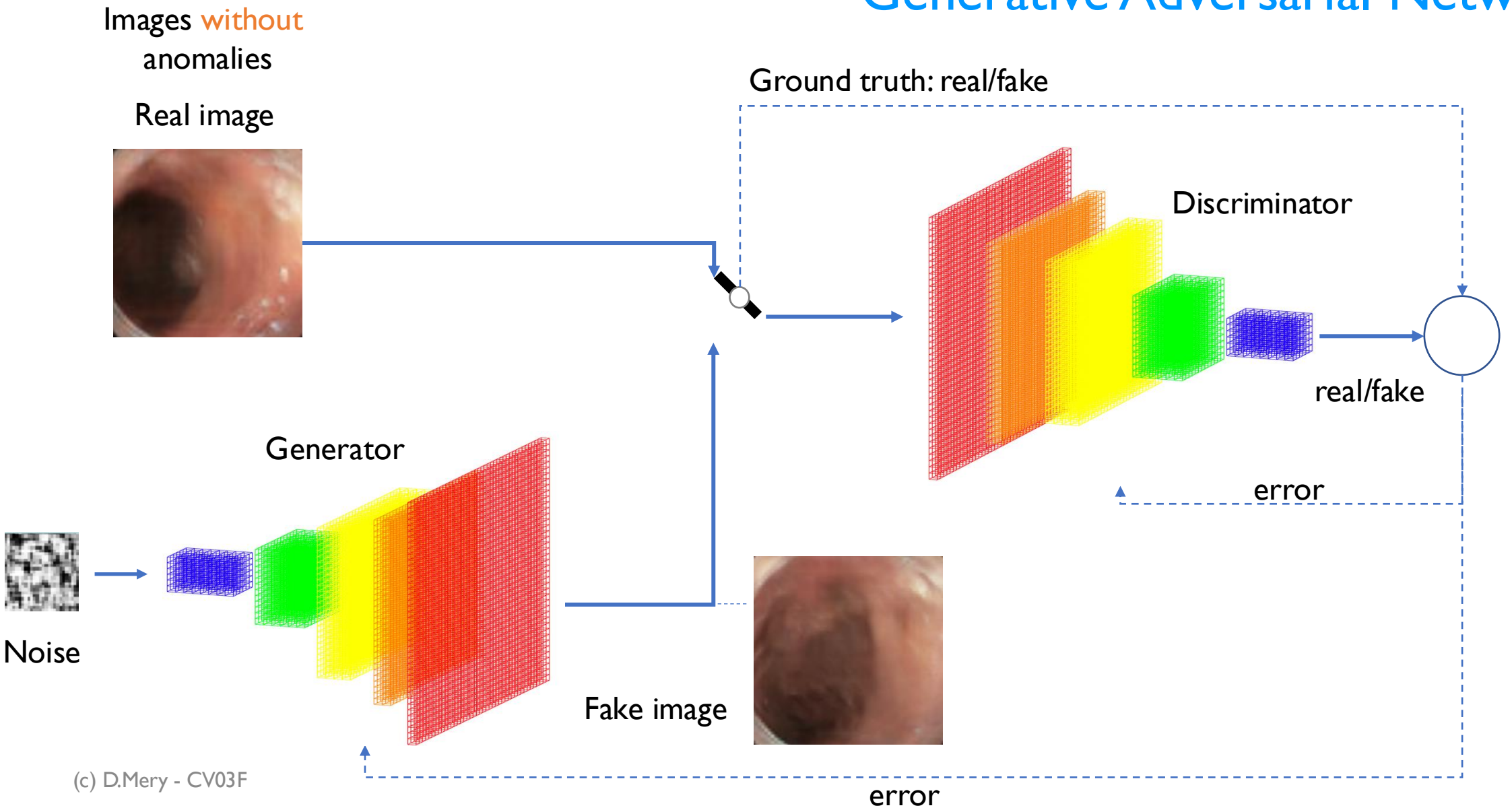
Anomaly Detection – Solution

Deep Learning Model: GAN
Generative Adversarial Network



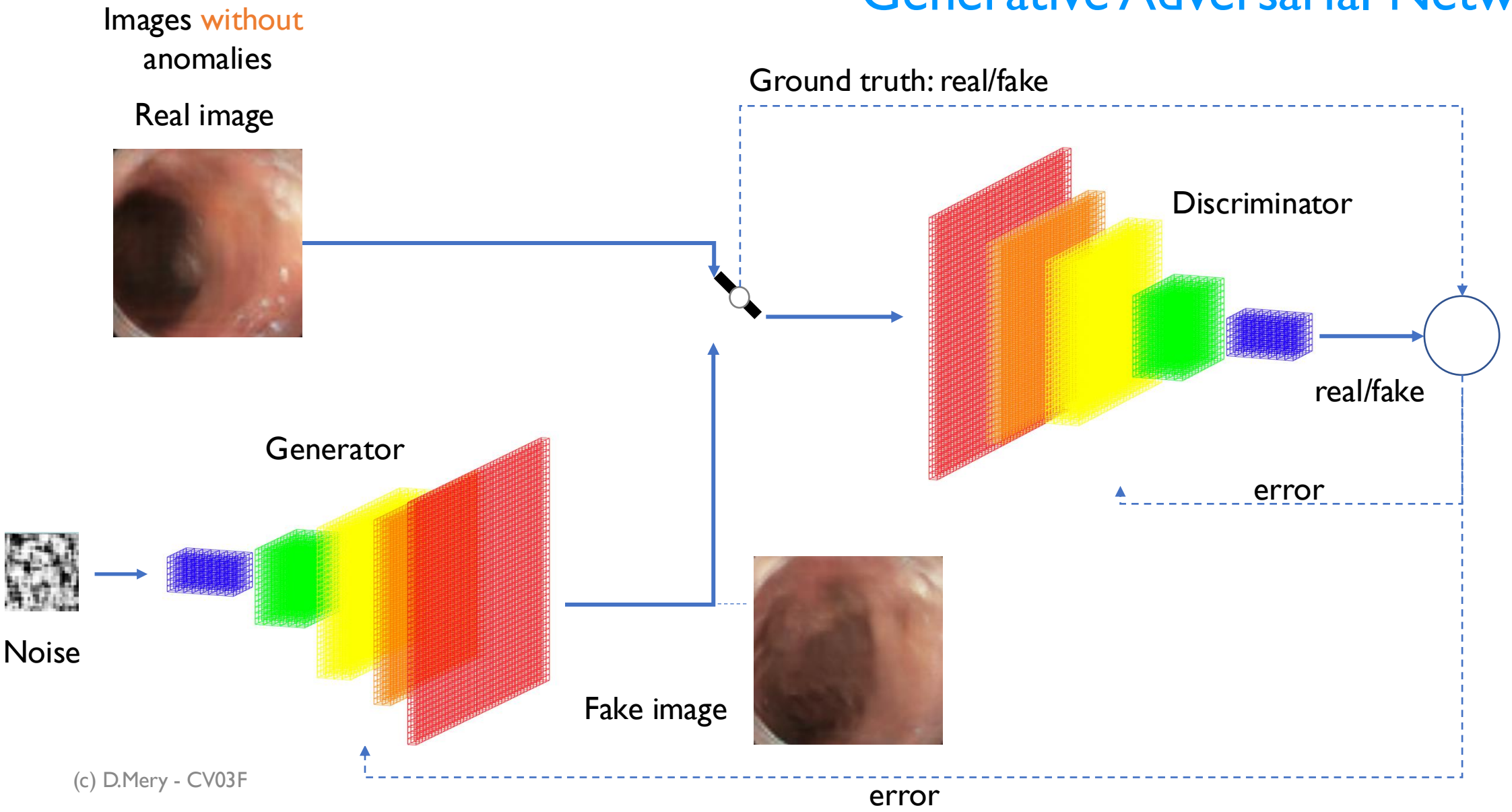
Anomaly Detection – Solution

Deep Learning Model: GAN
Generative Adversarial Network



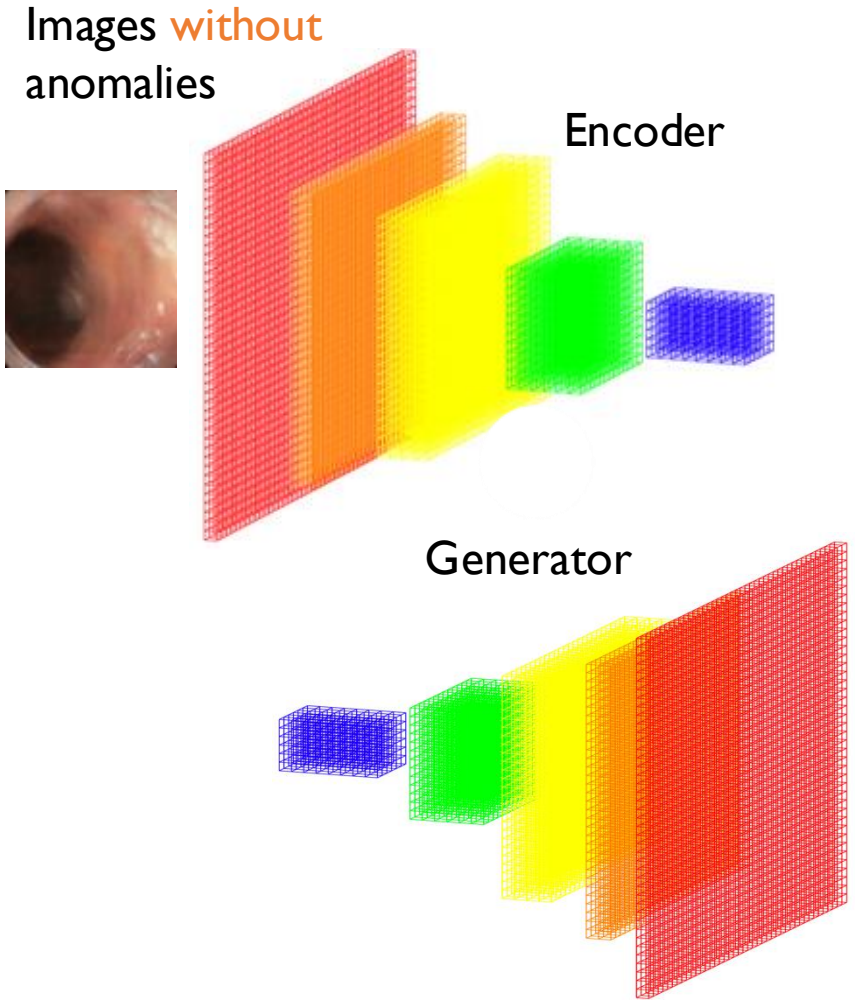
Anomaly Detection – Solution

Deep Learning Model: GAN
Generative Adversarial Network



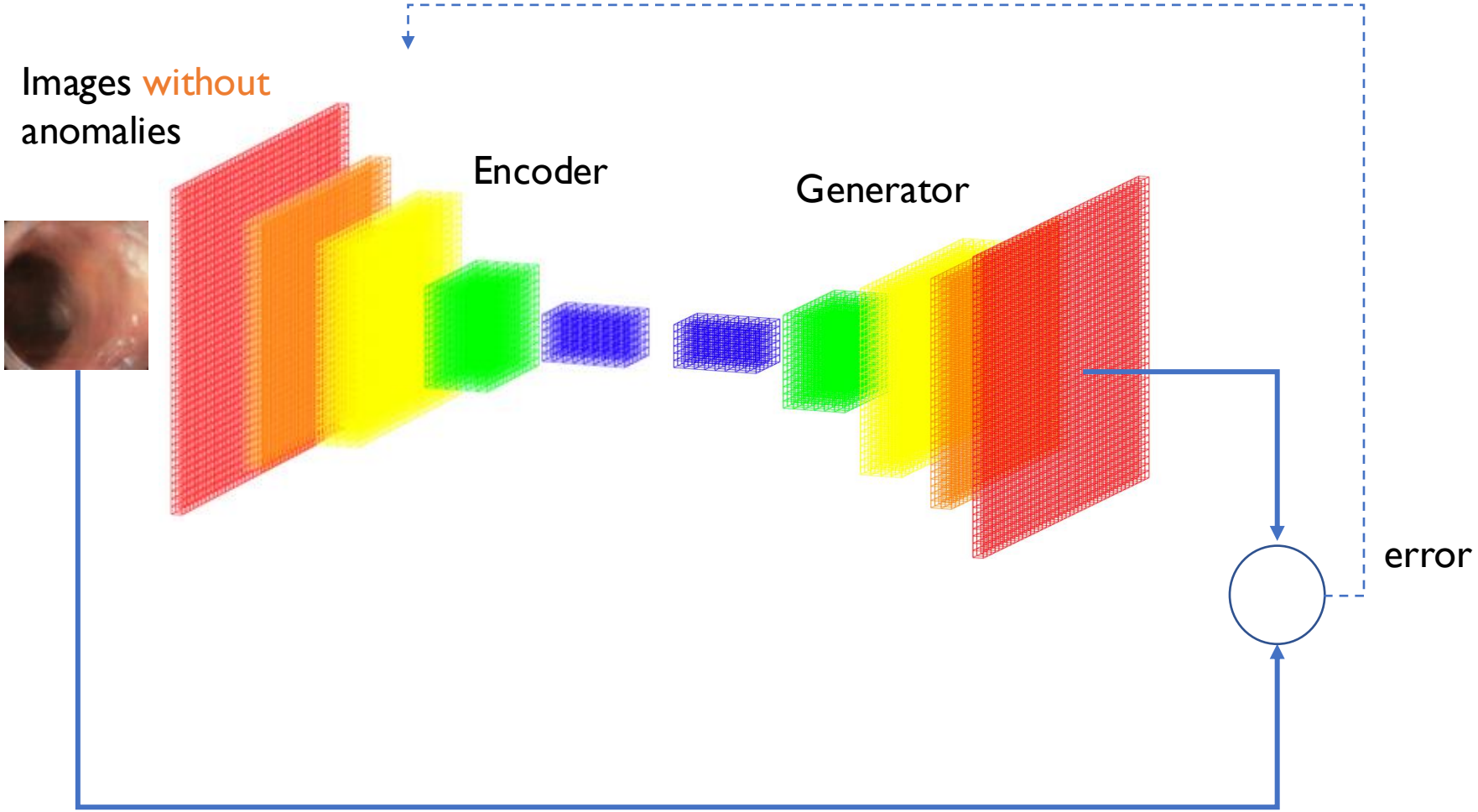
Anomaly Detection – Solution

Deep Learning Model: GAN
Generative Adversarial Network



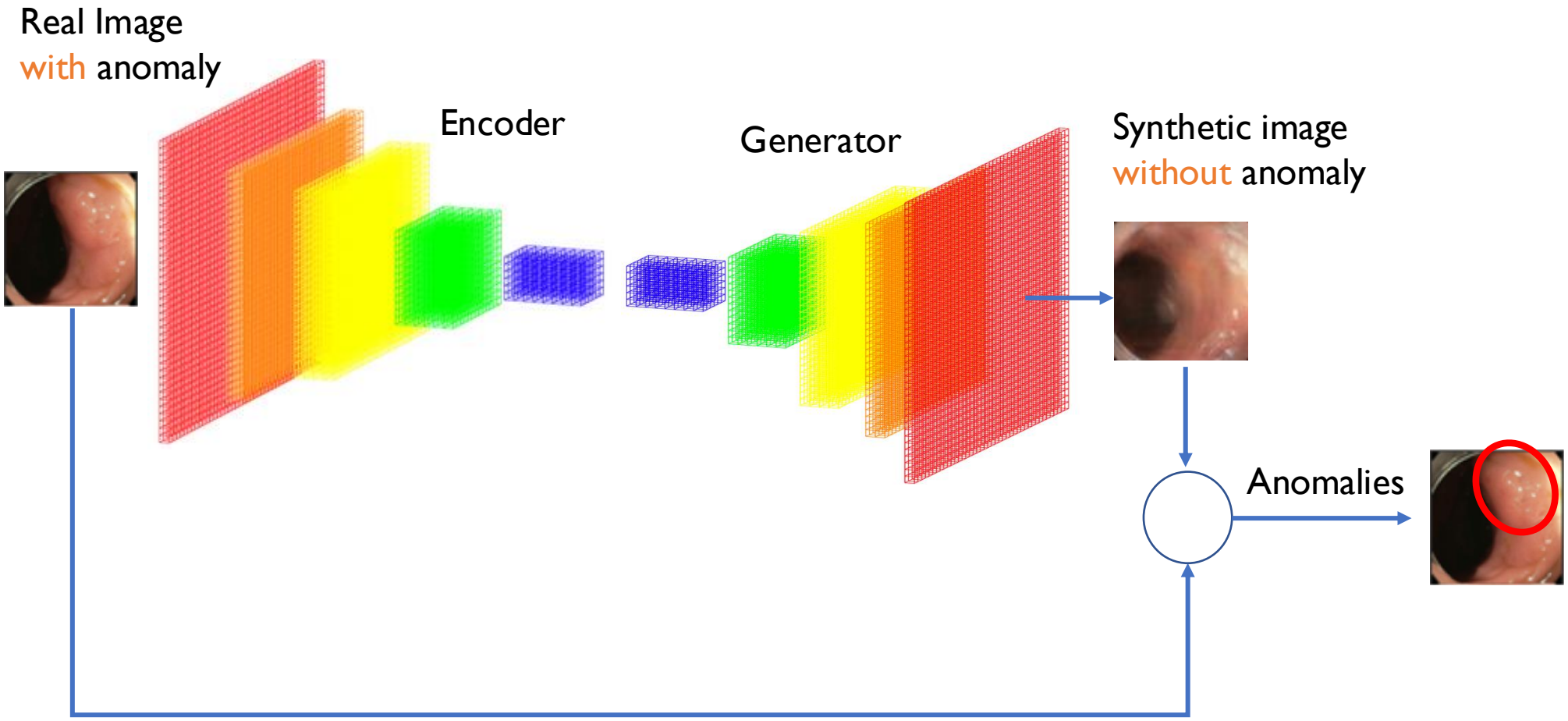
Anomaly Detection – Solution

Deep Learning Model: GAN
Generative Adversarial Network



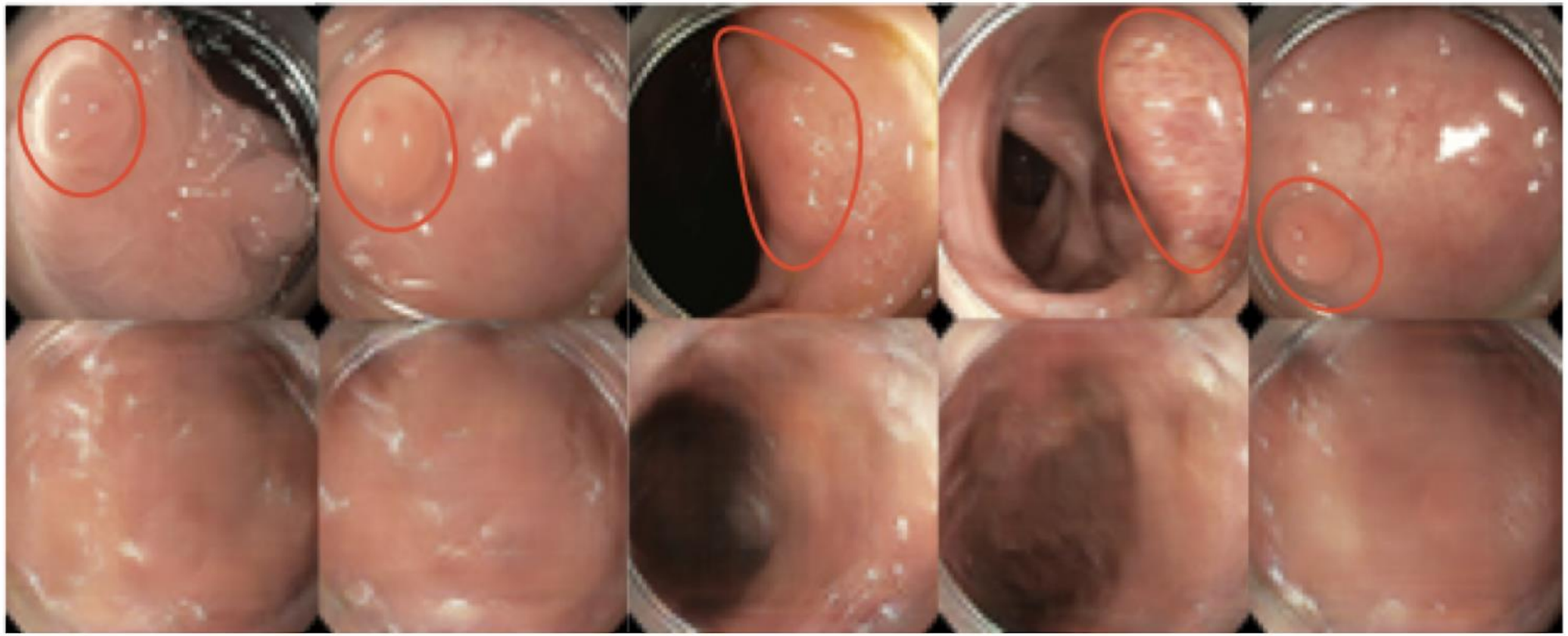
Anomaly Detection – Solution

Deep Learning Model: GAN
Generative Adversarial Network



Anomaly Detection – Example

Colonoscopy video frames



Liu, Yuyuan, et al. "Photoshopping colonoscopy video frames." *2020 IEEE 17th International Symposium on Biomedical Imaging (ISBI)*. IEEE, 2020.

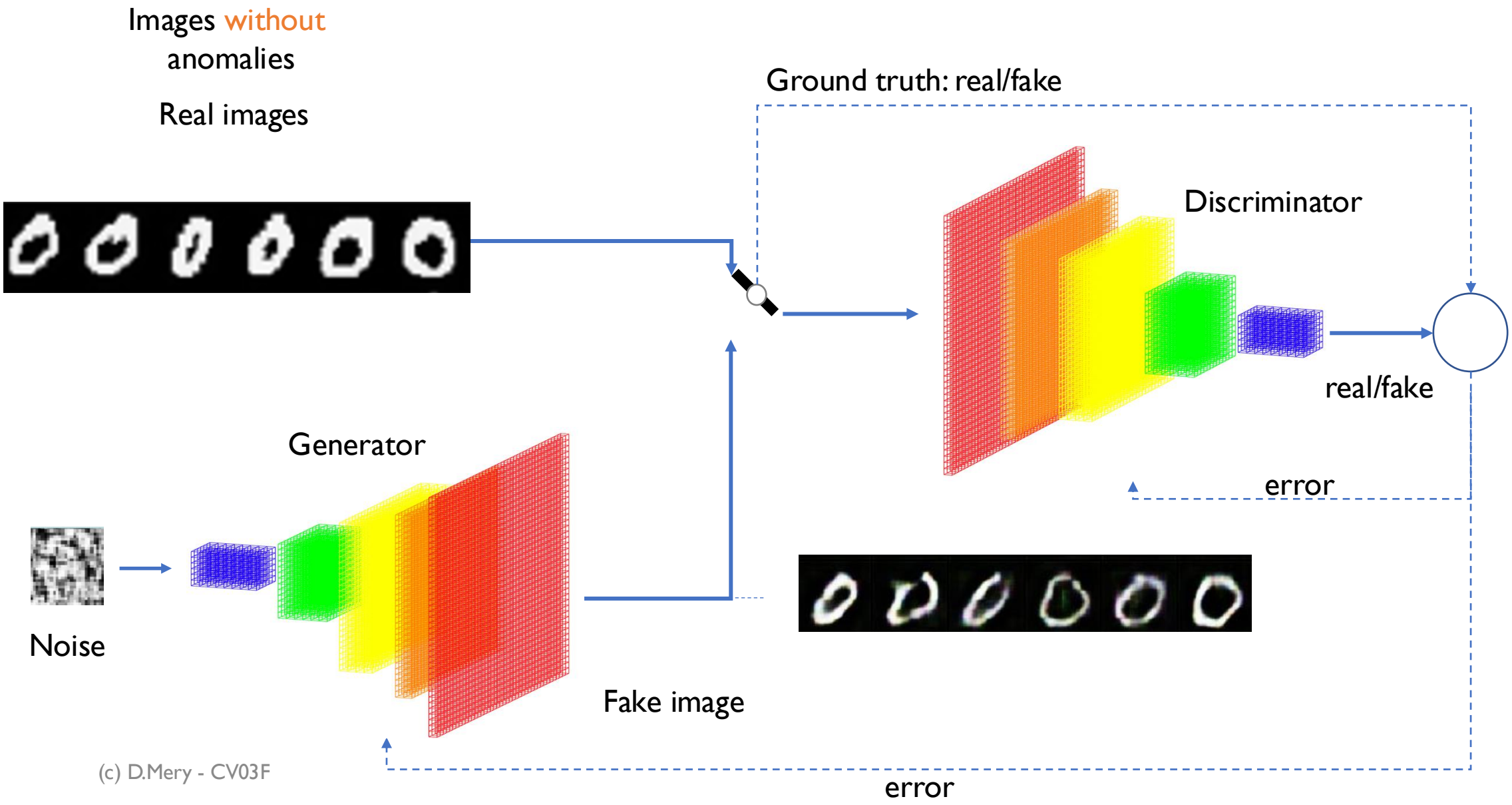
Anomaly Detection – Example



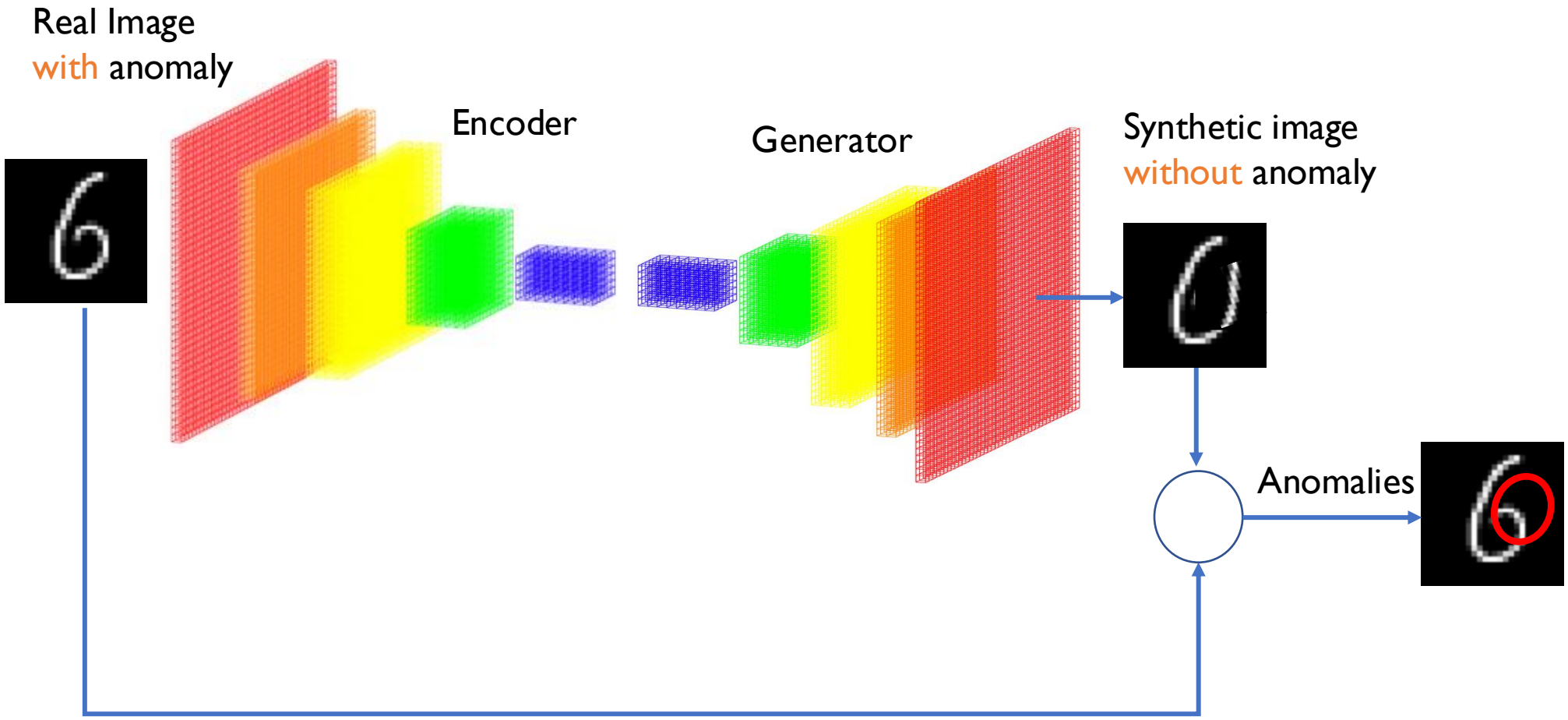
NORMAL

ANOMALY

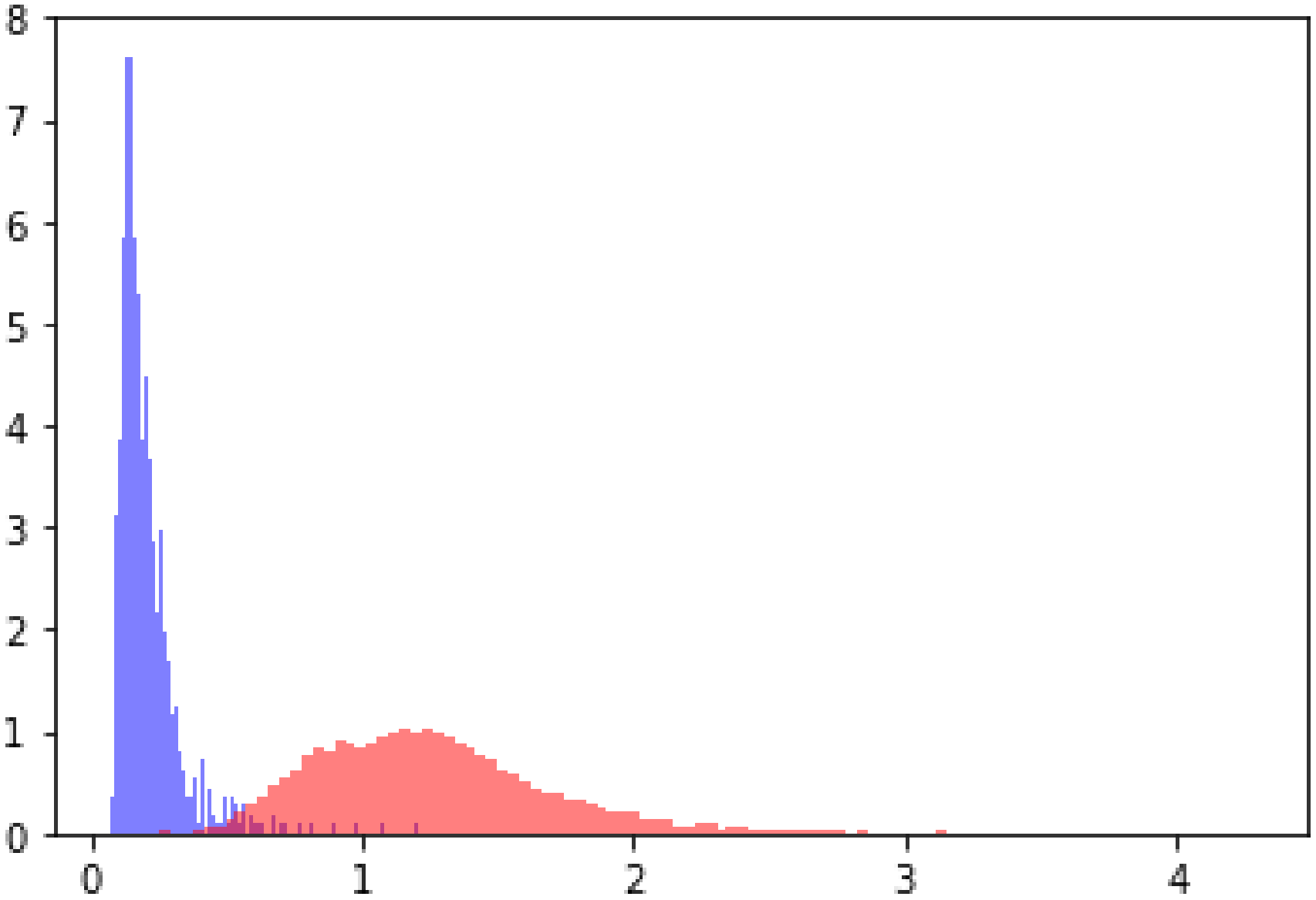
Anomaly Detection – Solution



Anomaly Detection – Solution

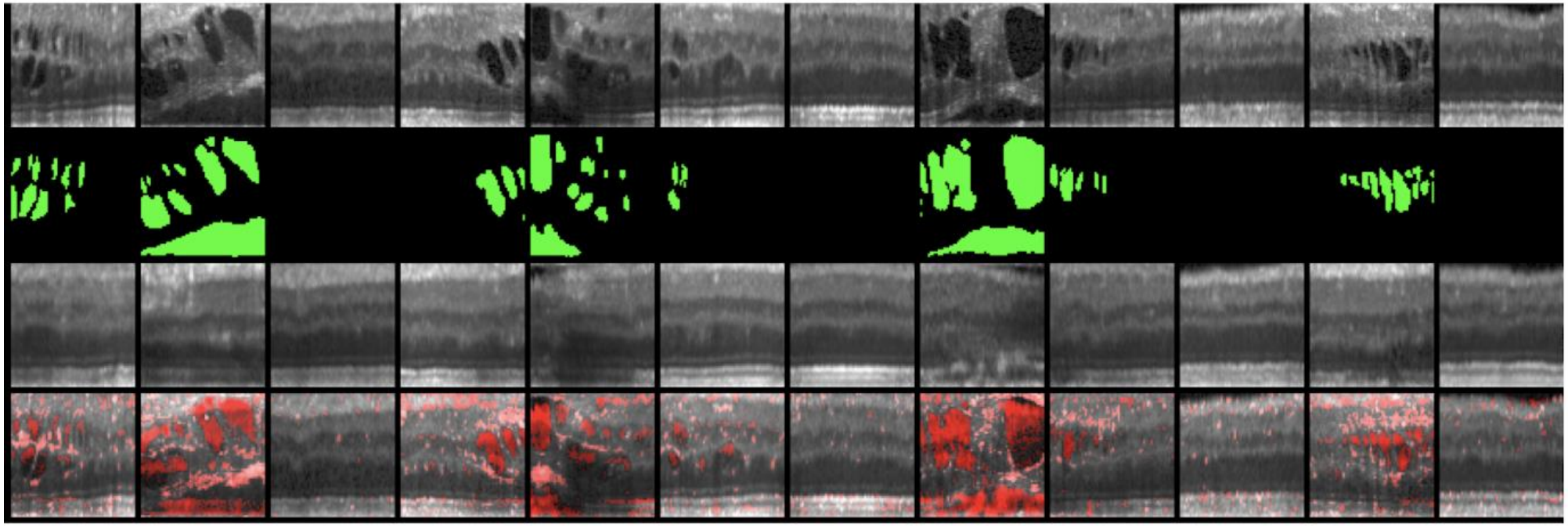


Anomaly Detection – Results 0 vs. the rest



Anomaly Detection – Example

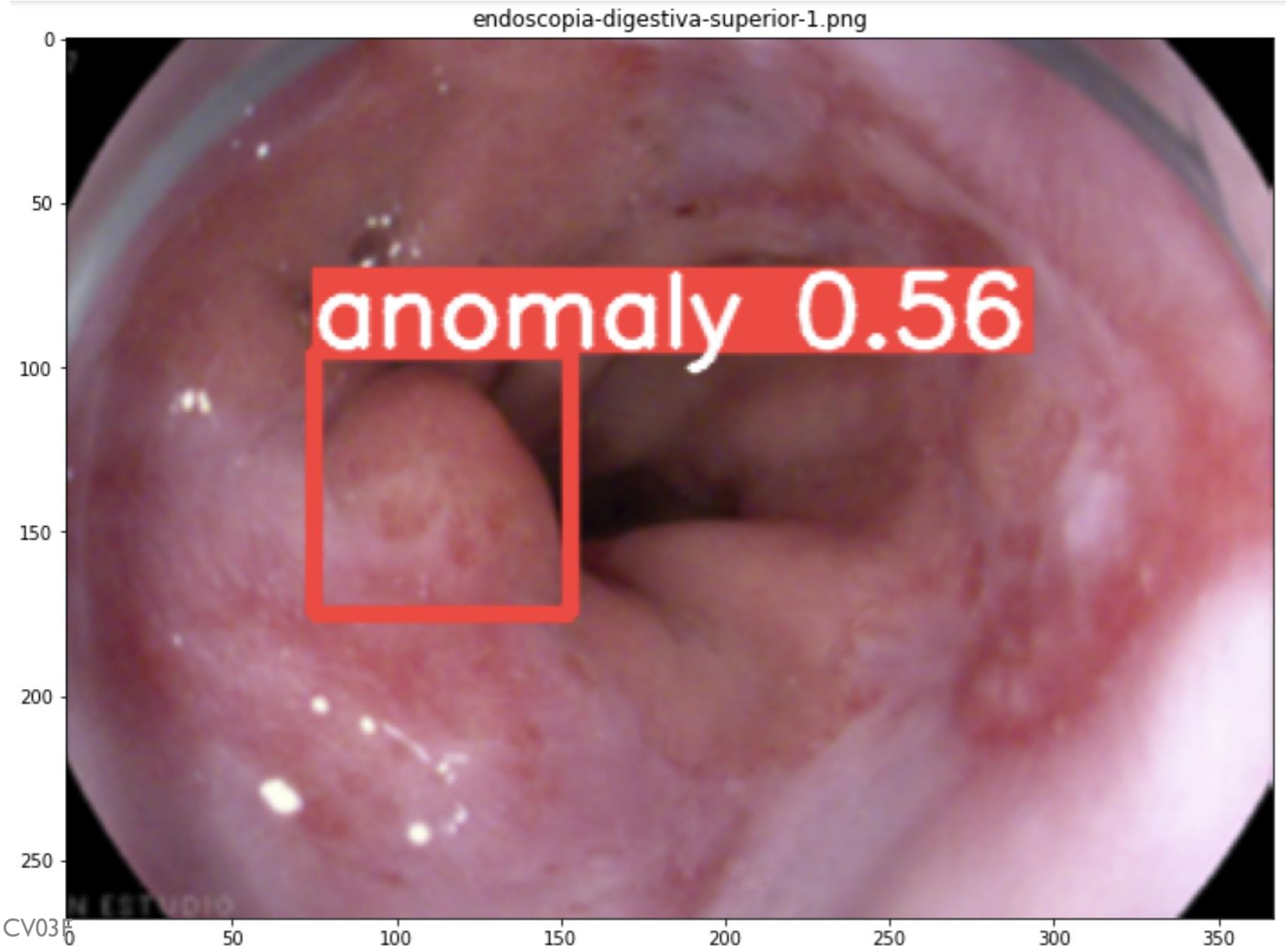
Retinal OCT images



Schlegl, Thomas, et al. "f-AnoGAN: Fast unsupervised anomaly detection with generative adversarial networks." Medical image analysis 54 (2019): 30-44.

Anomaly Detection – Bonus Track

YOLOv5



Créditos:
Javier Tramon